

Cofactor Exposure and the Missing–High Parity Kernel at the Critical Cutoff: Exact Matching Profiles, Restricted Native Grams, and a Determinant Reformulation

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Abstract

TPC–60 reduces restoration of the missing very-high Walsh face to a raw missing-mass estimate and an upper bound for the missing native grouping on the actual row-coefficient image. We identify both quantities exactly at the critical cutoff $y \asymp Q$ and one fixed nonzero shift.

Under the explicit identical-support specialization, each row m has a largest-prime cofactor exposure measure ν_m . Represented and missing raw masses are precisely $\nu_m(\mathcal{I}_m)$ and $\nu_m(\mathcal{I}_m^c)$. The optimal coefficient-uniform missing fraction is therefore the worst row defect, not an average cofactor density. We prove the fixed-shift cofactor ladder

$$j = j_0 + rt, \quad p = p_0 + mt,$$

and a Brun–Titchmarsh refinement. It yields a harmonic weighted defect certificate and shows why one omitted small cofactor can be more dangerous than its global cardinality suggests. Since the inherited interface leaves $\mathcal{I}_X$ as an arbitrary retained relation, it does not imply any nontrivial uniform missing-mass saving; this is a hypothesis-insufficiency theorem, not a counterexample to a more specific upstream carrier.

For the native map M_X , let $\mathbf{D}_M = \text{diag}(W_M(m))$. The optimal physical upper degree is

$$\kappa_M = \lambda_{\max} \left(\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2} \big|_{\text{Ran } \mathbf{D}_M} \right),$$

so $\|M_X \zeta\|^2 \leq \kappa_M \mathcal{D}_M(\zeta)$ sharply. On a nonempty missing carrier, fixed (i, p) incidence gives an exact unit diagonal and $\kappa_M = 1 + \lambda_{\max}(\Xi_{\neq p})$; all excess comes from unequal high primes. After a fixed row gauge, the off-diagonal entries are weighted two-affine-form Möbius correlations of $\mu(mj + h_0)\mu(m'j + h_0)$, equivalently sums on $m'pr - mp'r' = h_0(m' - m)$. This is the explicit missing-high parity kernel. We give graph, stable-rank, Schur-complement, and block certificates for its positive spectrum and a residual-level endpoint wedge. No bound $\kappa_M = X^{o(1)}$, missing-mass saving, parity improvement, prime-pair lower bound, or twin-prime consequence is asserted.

Keywords: fixed shift; largest-prime cofactor; weighted exposure; restricted Gram; Möbius correlation; parity kernel.

Convention. One nonzero physical shift h_0 , one direct residue/orbit slice, and one side of the inherited Walsh system are fixed unless an orthogonal direct sum is displayed. Every atomic Hilbert norm is the raw counting norm.

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1 Introduction

The critical-cutoff correction of TPC-59 reopens the geometric cofactor scale $r \asymp J$, but it does not say that the terminal retained-pair carrier meets the largest-prime cofactors actually exposed by the complete Walsh face. TPC-60 consequently splits that face into represented and missing atoms and writes their physical native outputs on one row-coefficient space. If A_X is the base plus represented-high map and M_X is the missing-high correction, the exact full map is

$$B_X = A_X + M_X. \quad (1)$$

Restoring the missing atoms is not monotone after grouping. TPC-60 proves the sharp pointwise certificate

$$\frac{\|B_X \zeta\|^2}{\|A_X \zeta\|^2} \geq \left(1 - \sqrt{\frac{d_{\text{miss}} \mathcal{D}_M(\zeta)}{\|A_X \zeta\|^2}}\right)_+^2, \quad (2)$$

but its ambient degree d_{miss} is far too expensive at the endpoint. The next task is therefore not another abstract lower angle. It is to determine, on the literal carrier,

1. how much raw high mass lies outside the retained cofactor relation;
2. how large the missing grouping is on the actual row-coefficient image.

This paper answers both questions at the level of exact objects and sharp finite-dimensional certificates. It does not estimate the final arithmetic constants. The distinction matters: replacing an unknown quantity by its correct formula is progress only if the formula produces new tests, new closures, or a narrower surviving target. Here it does all three.

1.1 From cofactor counts to exposure matching

On the very-high face, every target has the unique factorization

$$mj + h_0 = p_m(j)r_m(j), \quad p_m(j) = P^+(mj + h_0) > y. \quad (3)$$

For a row m , we retain the complete literal weight of all opposite rows and define the cofactor exposure $\nu_m(r)$ at those orbit times for which $r_m(j) = r$. Under identical Walsh and terminal supports, the retained set $\mathcal{I}_m = \{r : (m, r) \in \mathcal{I}_X\}$ captures exactly

$$W_R(m) = \nu_m(\mathcal{I}_m), \quad W_M(m) = \nu_m(\mathcal{I}_m^c). \quad (4)$$

Thus the problem is a weighted matching problem between a deterministic retained relation and a largest-prime exposure profile. Marginal knowledge of $\#\mathcal{I}_m$, or an average over rows, does not determine this match.

There is, however, new arithmetic structure inside one row. The terminal support forces $(m, r) = 1$. Fixing (m, r) in (3) gives

$$j = j_0 + rt, \quad p = p_0 + mt. \quad (5)$$

Consequently a fixed critical cofactor $r \asymp J$ occurs at only $O(1)$ orbit times. For smaller r , the relevant defect is harmonic: the candidate load is $1 + J/r$, with a logarithmic Brun–Titchmarsh improvement. A single omitted $r = 1$ may therefore have a length- J candidate ladder even though it is a negligible fraction of the global cofactor universe.

1.2 The optimal physical upper degree

The raw missing diagonal is a quadratic form on the row space:

$$\mathcal{D}_M(\zeta) = \langle \mathbf{D}_M \zeta, \zeta \rangle, \quad \mathbf{D}_M = \text{diag}(W_M(m)). \quad (6)$$

The least constant in

$$\|M_X \zeta\|^2 \leq \kappa \mathcal{D}_M(\zeta) \quad (7)$$

is

$$\kappa_M = \lambda_{\max} \left(\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2} |_{\text{Ran } \mathbf{D}_M} \right). \quad (8)$$

This is not the ambient native fiber degree. It is the exact operator norm after the atomic lift is restricted to the coupled physical row image.

The fixed- (i, p) singleton incidence makes each prime block an isometry relative to its raw diagonal. When $S_M \neq \{0\}$, normalization gives

$$\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2} = I + \Xi_{\neq p}, \quad \kappa_M = 1 + \lambda_{\max}(\Xi_{\neq p}). \quad (9)$$

All excess above one is therefore unequal-prime alias. The matrix further splits into same-residual determinant-ladder collisions and the new unequal-residual alias created by native erasure. Stable rank, weighted collision graphs, Schur complements, and dyadic block decompositions give rigorous ways either to certify a small positive spectrum or to kill that possibility before attempting a delicate number-theoretic proof.

1.3 The parity kernel is literal

The remaining Gram is not an unnamed “random-sign” object. Since $mj + h_0 = pr$ is squarefree,

$$-\mu(r) = \mu(mj + h_0). \quad (10)$$

The fixed factor $\mu(d_m)$ can be removed by a diagonal unitary gauge. For two distinct rows, the normalized Gram entry is then a weighted correlation of

$$\mu(mj + h_0) \mu(m'j + h_0) \quad (11)$$

on the simultaneous missing very-high support. Writing $mj + h_0 = pr$ and $m'j + h_0 = p'r'$, the common-orbit condition is exactly

$$m'pr - mp'r' = h_0(m' - m). \quad (12)$$

Thus the surviving route is an operator-norm problem for a truncated two-affine-form Möbius kernel, equivalently a signed determinant-surface sum. Known fixed-form or averaged cancellation cannot simply be inserted: the slopes range with X , the support includes a largest-prime cutoff and cofactor mismatch, and the desired quantity is a matrix norm rather than a single unrestricted average.

1.4 Main consequences and boundary

For $A_X \zeta \neq 0$, the exact replacement of (2) is

$$\boxed{\frac{\|B_X \zeta\|^2}{\|A_X \zeta\|^2} \geq \left(1 - \sqrt{\frac{\kappa_M \mathcal{D}_M(\zeta)}{\|A_X \zeta\|^2}} \right)_+^2}. \quad (13)$$

For a distinguished vector, κ_M may be replaced by its one-vector Rayleigh quotient; that conclusion remains pointwise. If

$$\kappa_M \leq X^{\nu_M+o(1)}, \quad \mathcal{D}_M(\zeta) \leq X^{-\sigma_M+o(1)}\mathcal{D}_{\text{ref}}(\zeta), \quad \|A_X\zeta\|^2 \geq X^{-\lambda_{\text{rep}}+o(1)}\mathcal{D}_{\text{ref}}(\zeta), \quad (14)$$

then the strict condition

$$\sigma_M > \nu_M + \lambda_{\text{rep}} \quad (15)$$

gives zero fixed-power reassembly loss in the same declared scope.

What is proved here is the exposure disintegration, the row ladder, the harmonic defect certificate, the optimal restricted Gram and its exact parity-kernel representation. What is not proved is a positive exposure match, a subpower bound for κ_M , or any lower square at one fixed shift. The result is L0/L1. It neither claims nor hints at an L2 parity advance, a prime-pair asymptotic, or a twin-prime consequence.

The paper is organized as follows. [Section 2](#) freezes the literal carrier. [Sections 3](#) to [5](#) treat the missing-mass side. [Sections 6](#) to [8](#) identify and test the restricted native Gram. The endpoint ledger and exact next gate appear in [sections 9](#) and [10](#).

2 The literal missing-high interface

We begin on the same one-side direct slice as TPC–53–60. The purpose of this section is to freeze the arithmetic carrier before defining either an exposure profile or a normalized Gram. None of the sets below may depend on the coefficient vector.

2.1 Scales, rows, and the critical cutoff

The inherited source row has the unique factorization

$$m = d_m \ell_m, \quad d_m \asymp D, \quad \ell_m \asymp L, \quad m \asymp Q = DL, \quad (16)$$

inside one declared row system. The orbit time satisfies $j \asymp J$, and at the endpoint

$$\begin{aligned} Q &= X^{267/400+o(1)}, & J &= X^{133/400+o(1)}, \\ D &= X^{10049/52500+o(1)}, & L &= X^{99979/210000+o(1)}. \end{aligned} \quad (17)$$

The physical shift $h_0 \neq 0$ is fixed. The equality support is squarefree and obeys

$$(d_m, mj + h_0) = (d_m, h_0) = 1. \quad (18)$$

Retain the support endpoints $D_{\pm}, M_{\pm}, J_{\pm}, N_{\pm}$ of TPC–59 and $\Delta_M = M_+ - M_-$. Throughout, the cutoff y is *very-high admissible*:

$$y > \max\{|h_0|, D_+, J_+, \Delta_M, \sqrt{N_+}\}, \quad \frac{D_+ N_+}{y} < D_- N_-. \quad (19)$$

In particular, $y = C_{\text{vh}}Q$ is allowed for a sufficiently large fixed support-dependent constant [\[7\]](#). The cutoff is a feasibility condition, not an energy estimate.

2.2 Recovered labels and raw amplitudes

Let $\mathcal{W}_H(y)$ be the complete nonzero very-high Walsh carrier on this slice. A coordinate $w \in \mathcal{W}_H(y)$ uniquely recovers

$$w = (m, j, \gamma), \quad n(w) = mj + h_0, \quad p(w) = P^+(n(w)) > y, \quad (20)$$

and, because $p(w)^2 > n(w)$,

$$r(w) = \frac{n(w)}{p(w)}, \quad s(w) = d_m r(w). \quad (21)$$

The native key is

$$i(w) = (L, \gamma, j) \quad (22)$$

on the displayed left slice. The right slice exchanges the two row systems and is added by orthogonal direct sum. Thus a fixed native key fixes the side, opposite row, and the same physical orbit time j .

Literal factorization supplies

$$b_w(\zeta) = \beta_w \zeta_{m(w)}, \quad \zeta \in \mathcal{K}_X, \quad (23)$$

where β_w retains the source shapes, static mask, opposite-row factor, orbit multiplier, four-factor profile, and shaped terminal weight, but not the bounded row coefficient ζ_m . It is the stripped TPC-45 coefficient: the physical high character is applied separately. No atom-dependent free phase is introduced.

The represented terminal census injects into $\mathcal{W}_H(y)$. Write

$$\mathcal{W}_H = \mathcal{W}_R \dot{\cup} \mathcal{W}_M \quad (24)$$

for its image and complement. The raw diagonals are

$$\mathcal{D}_E(\zeta) := \sum_{w \in \mathcal{W}_E} |\beta_w|^2 |\zeta_{m(w)}|^2, \quad E \in \{R, M, H\}, \quad (25)$$

so $\mathcal{D}_H = \mathcal{D}_R + \mathcal{D}_M$ exactly.

On squarefree support, $\mu(s) = \mu(d_m)\mu(r)$. The missing native map is therefore

$$(M_X \zeta)_i := - \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i}} \mu(s(w)) \beta_w \zeta_{m(w)}. \quad (26)$$

This is the same sign convention as TPC-58 and TPC-60; inserting another Möbius sign would change the physical map [6, 8].

2.3 The two singleton incidences

The very-high inequalities have two consequences used repeatedly. First, for fixed (i, p) , there is at most one high atom. Indeed, i fixes j ; if p meets rows m, m' , then $p \mid (m - m')j$. The alternative $p \mid j$ is excluded by $p > J_+$, and $|m - m'| \leq \Delta_M < p$ then gives $m = m'$; unique row factorization recovers the atom. Second, for fixed (i, m) , the target $mj + h_0$, hence p, r, s , is already determined. Consequently

$$\#\{w \in \mathcal{W}_H : (i(w), p(w)) = (i, p)\} \leq 1, \quad \#\{w \in \mathcal{W}_H : (i(w), m(w)) = (i, m)\} \leq 1. \quad (27)$$

These are coordinate incidences, not statements that a native output has only one prime or only one row.

2.4 When missing means cofactor mismatch

The exposure theorem below uses the following explicit specialization.

Definition 2.1 (Identical-support specialization). The Walsh and terminal constructions have identical row, orbit, opposite-row, terminal-prime, literal-mask, and profile supports, with no additional terminal-only pruning. If

$$\mathcal{I}_m := \{r : (m, r) \in \mathcal{I}_X\}, \quad (28)$$

then

$$w \in \mathcal{W}_M \iff r(w) \notin \mathcal{I}_{m(w)}. \quad (29)$$

TPC-60 proves this conditional support identity. Outside this specialization one must retain the full first-failure partition; the cofactor complement alone need not equal $\mathcal{W}_M$.

2.5 Six-item physical audit

The construction satisfies the required interface audit.

1. Every coefficient is the literal $\beta_w \zeta_{m(w)}$.
2. Every target is $mj + h_0$ for one fixed nonzero physical shift.
3. All atomic energies use raw counting measure, with no fiber average.
4. The row, largest prime, cofactor, residual label, and native key are canonically recovered; only identically inactive rows are removed.
5. The carriers $\mathcal{W}_H, \mathcal{W}_R, \mathcal{W}_M$ and $\mathcal{I}_X$ are fixed before ζ is chosen.
6. Exposure and Gram identities have zero structural cost, but every eventual reassembly loss remains in the strict 1/400 ledger.

3 Exact cofactor exposure on each physical row

We first isolate the part of the missing-high problem that is already diagonal in the inherited row coefficients. This must be done before any residual or native grouping: after grouping, a statement about omitted atomic mass is no longer monotone. The notation below retains the literal coefficient and raw counting normalization of TPC-53–TPC-60 [4, 5, 7, 8].

3.1 The realized cofactor measure

Work in the identical-support specialization of [definition 2.1](#). Thus a high Walsh atom is represented precisely when its recovered pair (m, r) belongs to the fixed retained relation $\mathcal{I}_X$. Let $\mathcal{M}_X$ be the declared source-row set and retain the notation

$$\mathcal{I}_m = \{r \in \mathbb{N} : (m, r) \in \mathcal{I}_X\}, \quad m \in \mathcal{M}_X, \quad (30)$$

from (28). The set $\mathcal{I}_m$ is fixed independently of the coefficient vector.

Every $w \in \mathcal{W}_H(y)$ has the unique recovered labels in (20)–(21). In particular,

$$n(w) = m(w)j(w) + h_0, \quad p(w) = P^+(n(w)) > y, \quad r(w) = \frac{n(w)}{p(w)}. \quad (31)$$

Literal factorization is $b_w(\zeta) = \beta_w \zeta_{m(w)}$, as in (23); β_w contains every fixed physical factor but not the bounded row coefficient ζ_m .

Definition 3.1 (Realized row cofactor measure). For $m \in \mathcal{M}_X$, define the finite positive measure on $\mathbb{N}$

$$\nu_m(r) := \sum_{\substack{w \in \mathcal{W}_H(y) \\ m(w)=m, r(w)=r}} |\beta_w|^2. \quad (32)$$

Write

$$W_H(m) := \sum_r \nu_m(r), \quad W_R(m) := \sum_{r \in \mathcal{I}_m} \nu_m(r), \quad W_M(m) := \sum_{r \notin \mathcal{I}_m} \nu_m(r). \quad (33)$$

Thus $W_H = W_R + W_M$ row by row.

The measure ν_m is not counting measure on possible cofactors. It counts only cofactors actually realized by the largest-prime factorization of $mj + h_0$, with the complete coefficient-free physical square attached.

Proposition 3.2 (Exact row disintegration). *For every row-coefficient vector ζ ,*

$$\mathcal{D}_H(\zeta) = \sum_m W_H(m) |\zeta_m|^2, \quad (34)$$

$$\mathcal{D}_R(\zeta) = \sum_m W_R(m) |\zeta_m|^2, \quad (35)$$

$$\mathcal{D}_M(\zeta) = \sum_m W_M(m) |\zeta_m|^2. \quad (36)$$

In particular, the represented/missing split is an orthogonal coordinate projection before residual or native grouping.

Proof. The sets $\mathcal{W}_R$ and $\mathcal{W}_M$ are the disjoint subsets of $\mathcal{W}_H$ on which respectively $r(w) \in \mathcal{I}_{m(w)}$ and $r(w) \notin \mathcal{I}_{m(w)}$. Substitute (23) into the three raw atomic squares and group nonnegative terms first by m , then by r . No coherent sum occurs in this calculation. $\square$

3.2 Sharp uniform and distinguished exposure constants

For every row with $W_H(m) > 0$, define

$$\varepsilon_m := \frac{W_M(m)}{W_H(m)}, \quad \rho_m := \frac{W_R(m)}{W_H(m)} = 1 - \varepsilon_m. \quad (37)$$

Theorem 3.3 (Sharp rowwise exposure minimax). *Assume the bounded coefficient class contains a nonzero multiple of every coordinate direction. If $\mathcal{D}_H$ is not identically zero, then*

$$\boxed{\sup_{\mathcal{D}_H(\zeta) > 0} \frac{\mathcal{D}_M(\zeta)}{\mathcal{D}_H(\zeta)} = \max_{m: W_H(m) > 0} \varepsilon_m,} \quad (38)$$

and

$$\boxed{\inf_{\mathcal{D}_H(\zeta) > 0} \frac{\mathcal{D}_R(\zeta)}{\mathcal{D}_H(\zeta)} = \min_{m: W_H(m) > 0} \rho_m = 1 - \max_{m: W_H(m) > 0} \varepsilon_m.} \quad (39)$$

Both identities are sharp at every finite X .

Proof. Whenever $\mathcal{D}_H(\zeta) > 0$,

$$\frac{\mathcal{D}_M(\zeta)}{\mathcal{D}_H(\zeta)} = \sum_{m: W_H(m) > 0} \frac{W_H(m) |\zeta_m|^2}{\mathcal{D}_H(\zeta)} \varepsilon_m. \quad (40)$$

This is a convex combination of the row defects. It is at most their maximum, and concentrating ζ on a maximizing row attains that maximum after a harmless scalar rescaling into the coefficient class. The represented identity follows from $\rho_m = 1 - \varepsilon_m$, or by the same argument with a minimum. $\square$

For one distinguished vector, the exact and generally weaker quantity is

$$\varepsilon_*(\zeta) := \frac{\mathcal{D}_M(\zeta)}{\mathcal{D}_H(\zeta)} = \sum_m \frac{W_H(m)|\zeta_m|^2}{\mathcal{D}_H(\zeta)} \varepsilon_m. \quad (41)$$

A favorable value of $\varepsilon_*(\zeta)$ does not imply the uniform bound in (38).

Corollary 3.4 (Exact missing-only-row kill test). *If one active row satisfies*

$$W_R(m_0) = 0 < W_M(m_0), \quad (42)$$

then the optimal uniform represented-mass constant is zero and the optimal uniform missing fraction is one.

3.3 What the inherited relation does not determine

The papers preceding TPC-61 call $\mathcal{I}_X$ the retained or supported pair relation, but impose no rowwise completion rule on it; arbitrary finite cofactor multiplicity is explicitly allowed [4, 5]. Consequently the displayed inherited hypotheses do not force

$$\mathcal{I}_m \cap \text{supp } \nu_m \neq \emptyset \quad \text{on every active row.} \quad (43)$$

Proposition 3.5 (Interface-insufficiency theorem). *Within the finite-relation hypothesis class stated in TPC-53–TPC-60, there is no universal constant $c > 0$, and no exponent $\sigma > 0$, for which either*

$$\mathcal{D}_R \geq c\mathcal{D}_H \quad \text{or} \quad \mathcal{D}_M \leq X^{-\sigma+o(1)}\mathcal{D}_H \quad (44)$$

follows from scale compatibility, finiteness of $\mathcal{I}_X$, and boundedness of the row coefficients alone.

Proof. Those assumptions do not exclude a retained relation which omits every realized cofactor on one otherwise active row. Such a row satisfies (42), and [corollary 3.4](#) gives both failures. This is a non-implication for the stated interface, not a claim that the upstream physical provenance necessarily chooses such a relation. $\square$

Example 3.6 (Average-row dilution does not repair a uniform defect). *Take any number of completed rows with $\varepsilon_m = 0$ and one active missing-only row with $\varepsilon_{m_0} = 1$. The unweighted average of the row defects can tend to zero as the number of completed rows grows, whereas (38) remains equal to one. The coefficient vector supported on m_0 detects the bad row exactly. Thus an average census can support a distinguished-vector statement only after the physical coefficient weights in (41) are retained.*

The result above closes only a proof architecture: it says that the opaque “retained relation” interface is insufficient. It is not a negative theorem about the actual upstream carrier. A genuine arithmetic advance must either reopen the provenance of $\mathcal{I}_X$, or directly estimate the realized weighted measures $\nu_m(I_m^c)$.

4 The fixed-shift cofactor ladder

We now use the literal equation to bound how much of one row can be captured by a specified missing cofactor. The resulting weight is harmonic in r , rather than uniform over the cofactor set.

4.1 Coprimality of a realized cofactor

Write every source row in its inherited unique form

$$m = d\ell, \quad d \in [D, 2D], \quad \ell \in [L, 2L] \cap \mathbb{P}, \quad (45)$$

where d is squarefree, $(d, h_0) = 1$, and $L > |h_0|$ for large X . On the common high carrier, the TPC-45 support gives $(d, r) = 1$ and squarefreeness of dr [2, 7, 8].

Lemma 4.1 (Full row-cofactor coprimality). *For every actual high atom,*

$$\boxed{(m, r) = 1.} \quad (46)$$

Moreover $(m, h_0) = 1$.

Proof. It remains only to exclude $\ell \mid r$. If this held, then $\ell \mid rp = mj + h_0$, while $\ell \mid mj$; hence $\ell \mid h_0$, contradicting $\ell > |h_0|$. Thus $(\ell, r) = 1$, and the inherited relation $(d, r) = 1$ proves (46). Finally $(d, h_0) = 1$ and $\ell > |h_0|$ imply $(m, h_0) = 1$. $\square$

4.2 Exact rowwise progression

Let $\mathcal{J}_X$ denote the fixed orbit slice. It is contained in an integer interval of diameter at most $C_J J$, where C_J depends only on the declared compact normalized orbit interval. Direct residue splitting may delete points from this interval and therefore only improves the upper bounds below.

Proposition 4.2 (Exact cofactor ladder). *Fix a source row m and a positive integer r which is realized by at least one high atom. There are integers j_0, p_0 such that every integral solution of*

$$rp - mj = h_0 \quad (47)$$

has the unique form

$$\boxed{j = j_0 + rt, \quad p = p_0 + mt, \quad t \in \mathbb{Z}.} \quad (48)$$

In addition,

$$p \equiv h_0 r^{-1} \pmod{m}, \quad (49)$$

and this is a reduced residue class modulo m .

Proof. By lemma 4.1, multiplication by m is invertible modulo r . Hence $mj \equiv -h_0 \pmod{r}$ has one residue class $j \equiv j_0 \pmod{r}$. Substitution of $j = j_0 + rt$ into (47) gives $p = p_0 + mt$, and every such pair is an integral solution. Conversely, subtracting two solutions gives $r(p - p') = m(j - j')$; coprimality of m, r recovers the same parameterization. Reducing the original equation modulo m proves (49). The residue is reduced because both r and h_0 are coprime to m . $\square$

For the next estimates, let

$$N_m(r) := \# \left\{ j \in \mathcal{J}_X : \begin{array}{l} p = (mj + h_0)/r \text{ is prime, } p > y, \\ p = P^+(mj + h_0), \text{ and there is a } \gamma \\ \text{for which the common high masks pass} \end{array} \right\}. \quad (50)$$

All conditions after integrality only delete points from the ladder.

Corollary 4.3 (Deterministic rowwise capacity). *Uniformly in every realized m, r ,*

$$\boxed{N_m(r) \leq 1 + C_J \frac{J}{r}.} \quad (51)$$

Proof. The admissible j 's lie in one residue class modulo r inside an interval of diameter at most $C_J J$. $\square$

4.3 The available one-dimensional sieve improvement

The prime coordinate in (48) lies in one reduced residue class modulo m . Brun–Titchmarsh therefore improves (51) by a logarithm when the ladder is long. We include the short-ladder case explicitly so that no expression with $\log(J/r) \leq 0$ is used.

Proposition 4.4 (Brun–Titchmarsh row bound). *There is a constant depending only on the fixed orbit interval such that*

$$N_m(r) \ll \frac{m}{\varphi(m)} \left(1 + \frac{J/r}{\log(2 + J/r)} \right). \quad (52)$$

On the inherited row support,

$$\frac{m}{\varphi(m)} = X^{o(1)}. \quad (53)$$

Thus the prime condition saves at most a logarithm in this coefficient-blind row census; it does not change its fixed-power exponent.

Proof. Put $T = J/r$. If T is bounded by a sufficiently large constant depending on C_J , (51) is $O(1)$, which is dominated by the right side of (52) because $m/\varphi(m) \geq 1$.

Suppose now that T exceeds that constant. The prime values $p = p_0 + mt$ lie in an interval of length at most $C_J m T + O(m)$ and in the reduced class (49). The interval form of the Brun–Titchmarsh theorem gives

$$\#\{p \text{ on this ladder}\} \ll \frac{mT}{\varphi(m) \log T}; \quad (54)$$

enlarging the interval by a fixed factor changes only the implied constant. The largest-prime and remaining physical conditions delete primes, so the same upper bound holds for $N_m(r)$. Combining the long and short cases and replacing $\log T$ by the uniformly positive $\log(2 + T)$ proves (52); see, for example, Montgomery and Vaughan [1].

Finally the standard maximal-order estimate $n/\varphi(n) \ll \log \log(3n)$, together with $m = X^{O(1)}$, proves (53). $\square$

Corollary 4.5 (Dyadic missing-cofactor census). *For a set $E_m(R) \subseteq [R, 2R] \cap \mathbb{N}$, the number of high orbit times on row m whose cofactor lies in $E_m(R)$ is at most*

$$\ll \frac{m}{\varphi(m)} |E_m(R)| \left(1 + \frac{J/R}{\log(2 + J/R)} \right). \quad (55)$$

In fixed-power notation, if

$$J = X^{\theta_J + o(1)}, \quad R = X^{\theta + o(1)}, \quad |E_m(R)| = X^{e(\theta) + o(1)}, \quad (56)$$

then the exponent in (55) is at most

$$e(\theta) + (\theta_J - \theta)_+. \quad (57)$$

Proof. Sum (52) over $r \in E_m(R)$, using $r \asymp R$. Equations (53) and (56) give the exponent formula. $\square$

The estimate is an upper capacity, not a lower bound for actual missing mass. Nevertheless it identifies the strongest conclusion obtainable from the fixed congruence and a one-dimensional upper-bound sieve alone. Any further fixed-power improvement must use the actual retained relation, the physical weights, a lower exposure estimate, or cancellation.

5 Harmonic defect, conditional savings, and the native wall

The row ladder shows that one omitted small cofactor is more dangerous than many omitted cofactors near the critical ceiling. We now turn this fact into a literal weighted estimate and audit its endpoint strength.

5.1 A weighted harmonic certificate

For each row and orbit time define the coefficient-free high weight

$$\Omega_m(j) := \sum_{\substack{w \in \mathcal{W}_H(y) \\ m(w)=m, j(w)=j}} |\beta_w|^2. \quad (58)$$

Every summand at a fixed (m, j) has the same recovered cofactor

$$r_m(j) := \frac{mj + h_0}{P^+(mj + h_0)} \quad (59)$$

whenever $P^+(mj + h_0) > y$. Different opposite-row atoms remain distinct in (58). Thus

$$W_H(m) = \sum_j \Omega_m(j), \quad W_M(m) = \sum_{\substack{j \\ r_m(j) \notin \mathcal{I}_m}} \Omega_m(j). \quad (60)$$

For an active row put

$$U_m := \max_j \Omega_m(j), \quad H_m^{\text{eff}} := \frac{W_H(m)}{U_m}. \quad (61)$$

The second quantity is a weighted effective number of exposed orbit times; it lies in $(0, \#\mathcal{J}_X]$. Let

$$E_m := \{r \in \mathbb{N} : r \leq N_+/y, r \notin \mathcal{I}_m\} \quad (62)$$

and define its rowwise sieve-harmonic capacity by

$$\mathfrak{h}_m(E_m) := \frac{m}{\varphi(m)} \sum_{r \in E_m} \left(1 + \frac{J/r}{\log(2 + J/r)}\right). \quad (63)$$

Including unrealized r 's in E_m only enlarges this upper certificate.

Theorem 5.1 (Literal harmonic-defect bound). *For every active row,*

$$\boxed{W_M(m) \ll U_m \mathfrak{h}_m(E_m)}, \quad (64)$$

and hence

$$\boxed{\varepsilon_m \ll \frac{\mathfrak{h}_m(E_m)}{H_m^{\text{eff}}}}. \quad (65)$$

Consequently, if a number $\epsilon_X \geq 0$ satisfies

$$\sup_{m: W_H(m) > 0} \frac{\mathfrak{h}_m(E_m)}{H_m^{\text{eff}}} \leq \epsilon_X, \quad (66)$$

then, with an absolute fixed-carrier constant,

$$\mathcal{D}_M(\zeta) \ll \epsilon_X \mathcal{D}_H(\zeta) \quad (67)$$

for every coefficient vector in the declared uniform class.

Proof. Group the second sum in (60) by the recovered cofactor. For fixed r , there are $N_m(r)$ eligible orbit times, and each aggregate weight is at most U_m . Therefore

$$W_M(m) \leq U_m \sum_{r \in E_m} N_m(r). \quad (68)$$

Apply proposition 4.4 and use (63); this proves (64). Division by $W_H(m) = U_m H_m^{\text{eff}}$ gives (65). The uniform conclusion follows from theorem 3.3. $\square$

The theorem is unconditional as an inequality, but a useful numerical value of its right side requires two arithmetic inputs: completion of the retained relation in harmonic capacity and a lower bound for the actual effective high exposure.

5.2 Dyadic exponent law

At the critical endpoint,

$$\theta_J := \frac{133}{400}, \quad J = X^{\theta_J + o(1)}, \quad \frac{N_+}{y} = X^{\theta_J + o(1)} \quad (69)$$

when $y = C_{\text{vh}} Q$ with the required fixed support constant [7]. Suppose a dyadic block $R = X^{\theta + o(1)}$, $0 \leq \theta \leq \theta_J$, satisfies uniformly on the claimed rows

$$|E_m \cap [R, 2R]| \leq X^{e(\theta) + o(1)}, \quad H_m^{\text{eff}} \geq X^{\eta_H + o(1)}. \quad (70)$$

For this block write

$$\mathcal{W}_{M,R} := \{w \in \mathcal{W}_M : R \leq r(w) < 2R\}, \quad (71)$$

$$W_{M,R}(m) := \sum_{\substack{R \leq r < 2R \\ r \notin \mathcal{I}_m}} \nu_m(r), \quad \mathcal{D}_{M,R}(\zeta) := \sum_m W_{M,R}(m) |\zeta_m|^2. \quad (72)$$

Corollary 5.2 (Dyadic cofactor-saving exponent). *The contribution of this block to the row defect obeys*

$$\frac{W_{M,R}(m)}{W_H(m)} \leq X^{-\sigma_{\text{cof}}(\theta) + o(1)}, \quad (73)$$

where

$$\sigma_{\text{cof}}(\theta) := \eta_H - e(\theta) - (\theta_J - \theta). \quad (74)$$

In the balanced-exposure case $\eta_H = \theta_J$, this simplifies to

$$\sigma_{\text{cof}}(\theta) = \theta - e(\theta). \quad (75)$$

The conclusion is useful only when the displayed saving is positive. An $O(\log X)$ dyadic union changes no fixed-power exponent.

Proof. By corollary 4.5, the harmonic numerator from the block has exponent at most $e(\theta) + \theta_J - \theta$. Divide by the effective-exposure lower bound in (70). Summing a logarithmic number of nonnegative block bounds contributes $X^{o(1)}$. $\square$

Example 5.3 (One small missing cofactor defeats global density). *Suppose all integers up to the critical ceiling are retained except $r = 1$. Its unweighted global missing fraction is $J^{-1+o(1)} = X^{-133/400+o(1)}$, but the row ladder is*

$$p = mj + h_0. \quad (76)$$

The support capacity has J candidates and Brun–Titchmarsh removes only a logarithm. Hence this cardinal statement certifies no positive fixed-power saving. If the actual high weight on a row is supported on the realized points of this ladder, then $\varepsilon_m = 1$. The current retained-relation interface contains no exposure hypothesis excluding that possibility.

Example 5.4 (Weight concentration defeats cofactor counting). *Even if the missing set has much smaller cardinality than the represented set, the exact measure can satisfy $\nu_m(\mathcal{I}_m) = 0 < \nu_m(\mathcal{I}_m^c)$ when all represented profile squares vanish and one missing profile square is nonzero. Less extremely, represented weights may be arbitrarily smaller than a missing weight. Thus cardinality can replace ν_m only after a two-sided physical-weight or effective- exposure estimate has been proved on the same carrier.*

These examples explain why the correct support statistic is harmonic and why even harmonic smallness must be divided by actual exposure. In particular, a proof based only on this certificate must cover every cofactor $r < X^{\sigma-o(1)}$ exactly, or supply an independent prime/weight saving on the omitted small-cofactor ladders, before claiming an $X^{-\sigma}$ missing-mass bound.

5.3 The residual wedge and the direct-native closure

The preceding exponent can now be compared with the grouping costs of TPC-60. Write

$$\lambda_L := \frac{99979}{210000}. \quad (77)$$

On a block $R = X^{\theta+o(1)}$, the ambient residual-key degree from the fixed-determinant ladder is

$$d_{\text{res}}(R) \leq X^{\lambda_L - \theta + o(1)}. \quad (78)$$

In the balanced-exposure scenario with missing-cardinality exponent $e(\theta)$, the phase-blind residual mass-degree exponent is therefore

$$\lambda_L - 2\theta + e(\theta). \quad (79)$$

At the critical block $\theta = 133/400$,

$$\lambda_L - \theta = \frac{15077}{105000}, \quad (80)$$

whereas

$$\boxed{2\theta - \lambda_L = \frac{39671}{210000} = 0.1889095\dots} \quad (81)$$

Thus the residual-key exponents themselves leave a genuine positive window. They do not close the physical map, because the latter also erases the residual label s .

To state the remaining requirement in the notation of the restricted-Gram section, factor the missing map on one block as

$$\mathcal{K}_X \xrightarrow{L_{M,R}} \ell^2(\mathcal{W}_{M,R}) \xrightarrow{C_{\text{res},R}} \ell^2(i, s) \xrightarrow{\mathbf{E}} \ell^2(i), \quad (82)$$

where $T_{M,R} := C_{\text{res},R} L_{M,R}$ is the residual map on the block and $\mathbf{E}$ is the canonical residual-label erasure. Define the blockwise restricted erasure constant

$$\beta_{M,R} := \left\| \mathbf{E}|_{\text{Ran } T_{M,R}} \right\|^2 = X^{\mu_{\text{erase}} + o(1)}. \quad (83)$$

This is the block analogue of β_M in [theorem 6.7](#); it is not the ambient number of residual labels at one native output. Likewise (78) gives

$$\kappa_{\text{res},R} := \sup_{\mathcal{D}_{M,R}(\zeta) > 0} \frac{\|T_{M,R}\zeta\|^2}{\mathcal{D}_{M,R}(\zeta)} \leq d_{\text{res}}(R) \leq X^{\lambda_L - \theta + o(1)}. \quad (84)$$

If the represented reference satisfies

$$\|A_X \zeta\|^2 \geq X^{-\lambda_{\text{rep}} + o(1)} \mathcal{D}_H(\zeta), \quad (85)$$

then the critical-block contribution has zero fixed-power reassembly cost under the strict condition

$$\boxed{e + \mu_{\text{erase}} + \lambda_{\text{rep}} < \frac{39671}{210000}}. \quad (86)$$

Indeed, the two-stage inequality

$$\|M_{X,R}\zeta\|^2 \leq \beta_{M,R}\kappa_{\text{res},R}\mathcal{D}_{M,R}(\zeta) \quad (87)$$

shows that the product of missing mass, residual degree, restricted erasure, and represented-reference loss is then $o(1)$. The reverse triangle certificate of TPC-60 yields the blockwise conclusion

$$\frac{\|(A_X + M_{X,R})\zeta\|^2}{\|A_X\zeta\|^2} = 1 + o(1). \quad (88)$$

For the full missing map, the same conclusion holds only if every other nonempty dyadic block is empty or satisfies its analogous strict estimate uniformly with a fixed positive margin. Since there are $O(\log X)$ blocks, those estimates give

$$\sum_R \|M_{X,R}\zeta\| = o(\|A_X\zeta\|), \quad (89)$$

and then $M_X = \sum_R M_{X,R}$ yields the full version of (88). A favorable critical block alone does not control missing mass on smaller cofactors. The condition is uniform or distinguished according to the scope of (85) and the cofactor-exposure hypotheses; the two scopes are not interchangeable.

By contrast, the only inherited direct native-degree bound has exponent

$$\mu_{\text{native}} = \frac{267}{400}. \quad (90)$$

Even the best nonempty pure-cardinality certificate at the critical scale, one missing cofactor with balanced exposure, saves at most $133/400$. Hence the direct-native phase-blind architecture has the fixed exponent deficit

$$\boxed{\frac{267}{400} - \frac{133}{400} = \frac{67}{200} > 0}. \quad (91)$$

It cannot prove reassembly by paying the full native degree. This is a route-specific closure, not a lower bound for the actual restricted norm and not an endpoint loss to be inserted into the ledger.

Claim boundary. The exact row measure, cofactor ladder, Brun-Titchmarsh estimate, harmonic certificate, and exponent audit are L0/L1 results at one fixed h_0 . No estimate in this section proves that the actual retained relation has a small harmonic complement, that every active row has polynomial high exposure, or that μ_{erase} satisfies (86). Consequently there is no L2 fixed-shift lower square, parity improvement, prime-pair asymptotic, or twin-prime implication. The positive conclusion is narrower and falsifiable: the residual wedge is open, the coefficient-blind direct native door is closed, and restricted Möbius erasure is the next arithmetic gate.

6 The restricted native Gram of the missing carrier

The ambient native degree from TPC-60 is the norm of a grouping operator on a free atomic space [8]. The physical missing vector does not range over that free space: every atom in one source row is driven by the same row coefficient. This section computes the optimal grouping constant on that actual coefficient image. It also separates the same-residual determinant-ladder collisions from the new collisions created when the residual label is erased.

6.1 Atomic, residual, and native maps

Let $\mathcal{W}_M$ be the fixed missing-high carrier, let $\mathcal{M}_X$ be the declared finite source-row set, and put

$$\mathcal{M}_M := \{m \in \mathcal{M}_X : \text{some } w \in \mathcal{W}_M \text{ satisfies } m(w) = m\}.$$

All maps below act on the original coefficient space $\mathcal{K}_X = \ell^2(\mathcal{M}_X)$; rows outside $\mathcal{M}_M$ remain zero directions of the missing diagonal rather than being deleted from the represented map. On an active missing atom w , write the literal coefficient as

$$b_w(\zeta) = \beta_w \zeta_{m(w)}, \quad m(w) \in \mathcal{M}_M, \quad (92)$$

and absorb the physical high-character sign and the residual Möbius gauge into

$$c_w := -\mu(s(w))\beta_w. \quad (93)$$

The active squarefree support gives $|c_w| = |\beta_w|$. Define the atomic lift

$$L_M : \mathcal{K}_X \longrightarrow \ell^2(\mathcal{W}_M), \quad (L_M \zeta)_w = c_w \zeta_{m(w)}. \quad (94)$$

Its Gram is diagonal:

$$\mathbf{D}_M := L_M^* L_M = \text{diag}(W_M(m))_{m \in \mathcal{M}_X}, \quad (95)$$

$$W_M(m) := \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2, \quad \mathcal{D}_M(\zeta) = \langle \mathbf{D}_M \zeta, \zeta \rangle. \quad (96)$$

Thus $\mathcal{D}_M$ is exactly the raw missing atomic diagonal, with no fiber normalization.

Let $i(w)$ be the native output key and $s(w)$ the residual label. The first grouping and the second erasure are

$$(T_M \zeta)_{i,s} := \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i, s(w)=s}} c_w \zeta_{m(w)}, \quad (97)$$

$$(\mathbf{E}z)_i := \sum_s z_{i,s}. \quad (98)$$

All sums use the active labels of the declared carrier. In particular, $\mathbf{E}$ does not adjoin ambient residual labels with zero physical coefficient.

Proposition 6.1 (Literal three-level factorization). *The physical missing map of TPC-60 satisfies*

$$\boxed{M_X = \mathbf{E} T_M}. \quad (99)$$

Moreover $\text{Ker } \mathbf{D}_M \subseteq \text{Ker } T_M \subseteq \text{Ker } M_X$.

Proof. At a native coordinate i , the right side of (99) is

$$\sum_s \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i, s(w)=s}} -\mu(s(w))\beta_w \zeta_{m(w)}.$$

This is exactly the sign-compatible missing-high contribution in the TPC-58/TPC-60 convention. If $\zeta \in \text{Ker } \mathbf{D}_M$, then every atom driven by a nonzero coordinate of ζ has zero fixed weight, so $L_M \zeta = 0$, and hence both grouped maps vanish. $\square$

Put

$$S_M := \text{Ran } \mathbf{D}_M = (\text{Ker } \mathbf{D}_M)^\perp. \quad (100)$$

Only the identically inactive rows are removed in S_M . In particular, one must not quotient by $\text{Ker } M_X$: a nonzero vector in that kernel is a genuine physical cancellation direction relative to the nonzero atomic norm $\mathcal{D}_M$.

All uniform suprema below are unchanged if they are restricted to the declared bounded coefficient cube, provided that cube contains a nonzero scalar multiple of every direction in $\mathcal{K}_X$. This follows from the degree-zero homogeneity of the Rayleigh quotients; it does not grant independent atom coefficients inside one source row.

6.2 The optimal effective native degree

Theorem 6.2 (Exact restricted native degree). *If $S_M \neq \{0\}$, define*

$$\kappa_M := \lambda_{\max} \left(\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2} |_{S_M} \right). \quad (101)$$

Then

$$\kappa_M = \sup_{\mathcal{D}_M(\zeta) > 0} \frac{\|M_X \zeta\|^2}{\mathcal{D}_M(\zeta)} \quad (102)$$

$$= \inf \{ \kappa \geq 0 : M_X^* M_X \preceq \kappa \mathbf{D}_M \}. \quad (103)$$

Consequently

$$\|M_X \zeta\|^2 \leq \kappa_M \mathcal{D}_M(\zeta) \quad (\zeta \in \mathcal{K}_X), \quad (104)$$

and κ_M is the least possible coefficient-uniform constant in (104). If $S_M = \{0\}$, then $M_X = 0$, and we set $\kappa_M = 0$.

Proof. For $\mathcal{D}_M(\zeta) > 0$, put $x = \mathbf{D}_M^{1/2} \zeta \in S_M$. Since M_X vanishes on $\text{Ker } \mathbf{D}_M$,

$$M_X \zeta = M_X \mathbf{D}_M^{\dagger/2} x.$$

The quotient in (102) is therefore the Rayleigh quotient of the positive operator in (101). Rayleigh–Ritz proves (102). A quadratic-form inequality for every ζ is equivalent to the Loewner inequality in (103), which proves optimality. $\square$

Let $C_M : \ell^2(\mathcal{W}_M) \rightarrow \ell^2(\mathcal{I}_{\text{out}})$ be the ambient native grouping

$$(C_M a)_i = \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i}} a_w, \quad d_{\text{miss}} := \max_i |\{w \in \mathcal{W}_M : i(w) = i\}|. \quad (105)$$

Then $M_X = C_M L_M$ after the unit gauges have been included in (93).

Corollary 6.3 (Compression improves the ambient degree). *One has*

$$\kappa_M \leq d_{\text{miss}}. \quad (106)$$

The inequality can be sharp, or the two sides can differ by an arbitrarily large factor.

Proof. The map

$$U_M := L_M \mathbf{D}_M^{\dagger/2} |_{S_M}$$

is an isometry because $U_M^* U_M = I_{S_M}$. Also $\|C_M\|^2 = d_{\text{miss}}$ by the exact fiberwise Cauchy–Schwarz calculation of TPC–60. Hence

$$\kappa_M = \|C_M U_M\|^2 \leq \|C_M\|^2 = d_{\text{miss}}.$$

Sharp examples are given in [examples 8.7](#) and [8.8](#). $\square$

6.3 Prime blocks and the unequal-prime alias operator

For $p > y$, let $\mathcal{W}_{M,p}$ be the missing atoms with terminal prime p , and let M_p be their contribution to the native map. Thus

$$M_X = \sum_{p>y} M_p. \quad (107)$$

Let

$$\mathbf{D}_p := \text{diag} \left(\sum_{\substack{w \in \mathcal{W}_{M,p} \\ m(w)=m}} |\beta_w|^2 \right)_{m \in \mathcal{M}_X}. \quad (108)$$

The critical-cutoff fixed- (i, p) singleton from TPC-59 and TPC-60 states that a fixed native output i and prime p support at most one active high atom [7, 8].

Theorem 6.4 (Prime-block isometry and exact alias Gram). *Assume $S_M \neq \{0\}$. On the literal fixed- h_0 carrier,*

$$M_p^* M_p = \mathbf{D}_p, \quad (109)$$

$$\sum_{p>y} \mathbf{D}_p = \mathbf{D}_M. \quad (110)$$

Define

$$\widehat{M}_p := M_p \mathbf{D}_M^{\dagger/2} |_{S_M}, \quad \Xi_{\neq p} := \sum_{\substack{p, q > y \\ p \neq q}} \widehat{M}_p^* \widehat{M}_q. \quad (111)$$

Then $\Xi_{\neq p}$ is Hermitian and

$$\sum_p \widehat{M}_p^* \widehat{M}_p = I_{S_M}, \quad (112)$$

$$\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2} |_{S_M} = I_{S_M} + \Xi_{\neq p}, \quad (113)$$

$$\kappa_M = \boxed{1 + \lambda_{\max}(\Xi_{\neq p})}. \quad (114)$$

Every nonzero matrix entry of $\Xi_{\neq p}$ is an unequal-prime alias at a common native output.

Proof. For a fixed p , the singleton property makes the summands at distinct atoms occupy distinct native coordinates. Therefore

$$\|M_p \zeta\|^2 = \sum_{w \in \mathcal{W}_{M,p}} |\beta_w|^2 |\zeta_{m(w)}|^2 = \langle \mathbf{D}_p \zeta, \zeta \rangle,$$

which proves (109). The prime carrier partition proves (110). Normalize this identity by $\mathbf{D}_M^{\dagger/2}$ to obtain (112). Expanding the square of $\sum_p \widehat{M}_p$ gives (113); the ordered terms (p, q) and (q, p) show that $\Xi_{\neq p}$ is Hermitian. Equation (114) follows from theorem 6.2. $\square$

The operator $\Xi_{\neq p}$ is generally indefinite; only $I + \Xi_{\neq p}$ is positive. This causes no loss in (114): adding the identity shifts every eigenvalue by exactly one, so $\lambda_{\max}(I + \Xi_{\neq p}) = 1 + \lambda_{\max}(\Xi_{\neq p})$ without any positivity assumption on $\Xi_{\neq p}$.

The diagonal of $\Xi_{\neq p}$ is zero. Indeed, i fixes the orbit time j , while i and a source row m fix $n = mj + h_0$, its unique prime $p = P^+(n) > y$, the cofactor $r = n/p$, and the residual label $s = d_m r$. Thus two different terminal primes from one source row cannot meet at the same native output. In particular, if $r_M := \dim S_M > 0$, the normalized native Gram has trace r_M .

6.4 Same-residual and unequal-residual spectra

Let $T_{M,p}$ denote the contribution of $\mathcal{W}_{M,p}$ to the residual map T_M , and set

$$\hat{T}_p := T_{M,p} \mathbf{D}_M^{\dagger/2}|_{S_M}. \quad (115)$$

Because the residual Hilbert space keeps (i, s) orthogonal, the cross-prime products in

$$\sum_{p \neq q} \hat{T}_p^* \hat{T}_q$$

retain only collisions with equal residual label.

Definition 6.5 (The two alias operators). Define

$$\Xi_{=s} := \sum_{\substack{p, q > y \\ p \neq q}} \hat{T}_p^* \hat{T}_q, \quad (116)$$

$$\Xi_{\neq s} := \Xi_{\neq p} - \Xi_{=s}. \quad (117)$$

Theorem 6.6 (Residual/native positive-spectrum split). *The two operators in [definition 6.5](#) are Hermitian, have zero diagonal, and satisfy*

$$\mathbf{D}_M^{\dagger/2} T_M^* T_M \mathbf{D}_M^{\dagger/2}|_{S_M} = I + \Xi_{=s}, \quad (118)$$

$$\mathbf{D}_M^{\dagger/2} M_X^* M_X \mathbf{D}_M^{\dagger/2}|_{S_M} = I + \Xi_{=s} + \Xi_{\neq s}. \quad (119)$$

Consequently, if

$$\kappa_{\text{res}} := \sup_{\mathcal{D}_M(\zeta) > 0} \frac{\|T_M \zeta\|^2}{\mathcal{D}_M(\zeta)}, \quad (120)$$

then

$$\kappa_{\text{res}} = 1 + \lambda_{\max}(\Xi_{=s}), \quad (121)$$

$$\boxed{\kappa_M \leq 1 + \lambda_{\max}(\Xi_{=s}) + \lambda_{\max}(\Xi_{\neq s})}. \quad (122)$$

Proof. The fixed- (i, s, p) singleton gives $T_{M,p}^* T_{M,p} = \mathbf{D}_p$. Expanding $T_M^* T_M$, normalizing, and using (110) proves (118). Subtract it from (113) to obtain (119). Rayleigh–Ritz gives (121), while the maximum-eigenvalue form of Weyl’s inequality gives (122). $\square$

The arithmetic roles of the two terms differ. The support of $\Xi_{=s}$ is governed by the determinant ladder

$$rp - dj\ell = h_0$$

inside one residual label. The operator $\Xi_{\neq s}$ is created only after s is erased; its terms share the native output i but have different residual labels. Thus the residual ladder does not by itself control $\Xi_{\neq s}$. Also, the present upper-norm problem needs only the positive spectra of these signed operators. Bounding their full absolute norms is sufficient but can discard favorable cancellation.

6.5 The quotient on the actual residual image

Put

$$\mathbf{R}_M := T_M^* T_M, \quad \mathbf{N}_M := T_M^* \mathbf{E}^* \mathbf{E} T_M = M_X^* M_X, \quad S_R := \text{Ran } \mathbf{R}_M = (\text{Ker } T_M)^\perp, \quad (123)$$

and let

$$\mathcal{E}_M := \text{Ran } T_M \quad (124)$$

be the actual residual coefficient image.

Theorem 6.7 (Exact restricted erasure norm). *If $\mathcal{E}_M \neq \{0\}$, then*

$$\beta_M := \sup_{0 \neq z \in \mathcal{E}_M} \frac{\|\mathbf{E}z\|^2}{\|z\|^2} \quad (125)$$

$$= \lambda_{\max}(\mathbf{R}_M^{\dagger/2} \mathbf{N}_M \mathbf{R}_M^{\dagger/2}|_{S_R}). \quad (126)$$

Moreover,

$$\boxed{\kappa_M \leq \kappa_{\text{res}} \beta_M}. \quad (127)$$

Proof. The polar-normalized map

$$V := T_M \mathbf{R}_M^{\dagger/2}|_{S_R} : S_R \longrightarrow \mathcal{E}_M$$

is unitary. Conjugating $\mathbf{E}^* \mathbf{E}|_{\mathcal{E}_M}$ by V gives the operator in (126). Finally,

$$\|M_X \zeta\|^2 \leq \beta_M \|T_M \zeta\|^2 \leq \beta_M \kappa_{\text{res}} \mathcal{D}_M(\zeta),$$

and optimality in theorem 6.2 proves (127). $\square$

For comparison with the coefficient-blind ladder bounds, put

$$d_{\text{res}} := \max_{i,s} |\{w \in \mathcal{W}_M : i(w) = i, s(w) = s\}|, \quad (128)$$

$$k_{\text{erase}} := \max_i |\{s : (i, s) \text{ is active in the missing residual carrier}\}|. \quad (129)$$

The two ambient grouping maps have squared norms d_{res} and k_{erase} , respectively. Compression to the physical images therefore gives

$$\boxed{\kappa_{\text{res}} \leq d_{\text{res}}, \quad \beta_M \leq k_{\text{erase}}}. \quad (130)$$

Neither reverse comparison is valid: the common row coefficients and the literal phases may make the restricted norms much smaller.

The ambient erasure norm is the largest number of active residual labels at one i , but β_M is its compression to $\mathcal{E}_M$. These need not be comparable from below. Likewise, the product in (127) is a useful certificate, not an identity in general; the one-stage constant κ_M remains optimal.

Remark 6.8 (Block notation and the residual Möbius gauge). On a dyadic cofactor block R , the quantity $\beta_{M,R}$ in (83) is precisely the specialization of β_M to the corresponding atomic and residual maps. If the residual Möbius sign is instead stored in a final signed erasure C_μ , rather than in the residual input as in (93), the two descriptions are conjugate by the diagonal unitary

$$(U_\mu z)_{i,s} = \mu(s) z_{i,s}.$$

Thus the two gauges have the same restricted norm. They describe one gate and must not both be charged in the endpoint ledger.

6.6 Uniform and distinguished quantifiers

For one coefficient vector with $\mathcal{D}_M(\zeta) > 0$, define

$$\kappa_M(\zeta) := \frac{\|M_X \zeta\|^2}{\mathcal{D}_M(\zeta)}. \quad (131)$$

Proposition 6.9 (Pointwise Rayleigh identity). *Let $x = \mathbf{D}_M^{1/2}\zeta$. Then*

$$\boxed{\kappa_M(\zeta) = \frac{\langle (I + \Xi_{\neq p})x, x \rangle}{\|x\|^2}, \quad 0 \leq \kappa_M(\zeta) \leq \kappa_M.} \quad (132)$$

The pointwise value may be zero even when the uniform constant is large.

Proof. Insert $x = \mathbf{D}_M^{1/2}\zeta$ into (113). Positivity gives the lower bound and theorem 6.2 gives the upper bound. $\square$

For the represented map A_X and every vector with $A_X\zeta \neq 0$, the reverse triangle inequality now gives the optimal mass-to-native certificate available from $\mathcal{D}_M$:

$$\begin{aligned} \frac{\|(A_X + M_X)\zeta\|^2}{\|A_X\zeta\|^2} &\geq \left(1 - \sqrt{\frac{\kappa_M \mathcal{D}_M(\zeta)}{\|A_X\zeta\|^2}}\right)_+^2 \\ &\geq \left(1 - \sqrt{\frac{d_{\text{miss}} \mathcal{D}_M(\zeta)}{\|A_X\zeta\|^2}}\right)_+^2. \end{aligned} \quad (133)$$

For one distinguished physical vector, κ_M in the first line can be replaced by the exact number $\kappa_M(\zeta)$. This replacement is pointwise only: a small Rayleigh quotient for one vector does not imply the Loewner inequality (103).

7 The literal parity kernel

The normalized Gram in section 6 is exact, but its off-diagonal operator still appears there as an unequal-prime alias. We now calibrate every sign in that operator. The result is a weighted correlation of the two affine values

$$mj + h_0 \quad \text{and} \quad m'j + h_0$$

on the actual missing carrier. No randomness hypothesis for the Möbius function is made, and no estimate for the resulting operator norm is asserted.

7.1 Row gauge and exact sign calibration

Retain the active row space $S_M = \text{Ran } \mathbf{D}_M$ and the unequal-prime alias operator $\Xi_{\neq p}$ from (111). Let

$$U_d : S_M \longrightarrow S_M, \quad (U_d z)_m := \mu(d_m)z_m. \quad (134)$$

The squarefree support and $(d_m, h_0) = 1$ make U_d a diagonal self-adjoint unitary. Since $\mathbf{D}_M$ is diagonal in the same row basis, U_d commutes with every power and Moore–Penrose power of $\mathbf{D}_M$.

For two active rows define the literal common-output form

$$\mathfrak{C}_{m,m'} := \sum_{\substack{w, w' \in \mathcal{W}_M \\ m(w)=m, \, m(w')=m' \\ i(w)=i(w')}} \mu(s(w))\mu(s(w'))\overline{\beta_w}\beta_{w'}. \quad (135)$$

For $m = m'$, the fixed- (i, m) singleton in (27) gives $\mathfrak{C}_{m,m} = W_M(m)$. For $m \neq m'$, the same expression is the off-diagonal entry of $M_X^* M_X$.

Proposition 7.1 (Exact row-gauge calibration). *Let $m \neq m'$ be active missing rows. Then*

$$(U_d \Xi_{\neq p} U_d)_{m,m'} = \frac{1}{\sqrt{W_M(m)W_M(m')}} \sum_{\substack{w,w' \in \mathcal{W}_M \\ m(w)=m, m(w')=m' \\ i(w)=i(w')}} \mu(n(w))\mu(n(w'))\overline{\beta_w}\beta_{w'}. \quad (136)$$

The diagonal of $U_d \Xi_{\neq p} U_d$ is zero, and

$$\lambda_{\max}(\Xi_{\neq p}) = \lambda_{\max}(U_d \Xi_{\neq p} U_d). \quad (137)$$

Consequently

$$\kappa_M = 1 + \lambda_{\max}(U_d \Xi_{\neq p} U_d). \quad (138)$$

Proof. On every high atom,

$$n(w) = p(w)r(w)$$

is squarefree, $p(w)$ is prime, and $(p(w), r(w)) = 1$. Hence

$$\mu(r(w)) = -\mu(n(w)), \quad \mu(s(w)) = \mu(d_{m(w)})\mu(r(w)) = -\mu(d_{m(w)})\mu(n(w)). \quad (139)$$

The two minus signs cancel in a Gram product. Conjugating its row indices by U_d removes the remaining fixed factors $\mu(d_m)\mu(d_{m'})$, leaving exactly (136). The normalization by $W_M(m)^{-1/2}$ is the action of $\mathbf{D}_M^{\dagger/2}$ on the active row basis. The diagonal statement was proved after (113); it also follows directly because one pair (i, m) determines one target and one very-high prime. Finally, unitary conjugation preserves eigenvalues, which gives (137); combine this with (114). $\square$

The source-divisor sign has therefore been removed exactly; it has not been discarded statistically. The remaining arithmetic sign is the product of the Möbius values of two actual affine targets. This is the parity-bearing part of the native alias Gram.

7.2 The direct-cell two-affine formula

On the displayed left direct cell, write a high atom as $w = (m, j, \gamma)$, extending its coefficient by zero when the declared literal support fails. The factor dictionary of TPC-53 gives the exact coefficient-free separation

$$\beta_{m,j,\gamma} = a_m(j)K_{m\gamma}q_\gamma(j), \quad (140)$$

where $a_m(j)$ contains the source and terminal shapes, $q_\gamma(j)$ contains the opposite-row shape and its fixed output phase, and $K_{m\gamma} \in \{0, 1\}$ is the literal static mask. The fixed residue/orbit phase may be placed in either factor; the product in (140) is the literal coefficient.

Let

$$\mathbf{m}_m(j) := \mathbf{1}_{\{(m,j,\gamma) \in \mathcal{W}_M \text{ for some } \gamma\}}, \quad (141)$$

where coefficient zeros caused only by $K_{m\gamma}$ or $q_\gamma(j)$ remain in the factors rather than in the indicator. In the identical-support specialization,

$$\mathbf{m}_m(j) = 1 \iff P^+(mj + h_0) > y, \quad r_m(j) \notin \mathcal{I}_m, \quad (142)$$

where $r_m(j)$ is defined in (59), and all common high-support tests are understood. Outside that specialization one must replace the last test by the full first-failure signature.

Put

$$\Gamma_{m,m'}(j) := \sum_{\gamma} K_{m\gamma}K_{m'\gamma}|q_\gamma(j)|^2. \quad (143)$$

Corollary 7.2 (Literal two-affine Möbius kernel). *For two distinct active rows m, m' ,*

$$\boxed{(U_d \Xi_{\neq p} U_d)_{m, m'} = \frac{1}{\sqrt{W_M(m) W_M(m')}} \sum_j \mathbf{m}_m(j) \mathbf{m}_{m'}(j) \times \mu(mj + h_0) \mu(m'j + h_0) \overline{a_m(j)} a_{m'}(j) \Gamma_{m, m'}(j).} \quad (144)$$

The sum is over the one fixed physical orbit slice. The right direct slice has the symmetric formula, and the two sides are joined only by orthogonal direct sum.

Proof. A common native output on the left is precisely a common pair (γ, j) . Substitute (140) into (136), sum first over γ , and use (143). The missing and high tests depend on m, j , not on the surviving opposite coordinate, in the identical-support specialization. $\square$

Equation (144) is an identity, not a Chowla estimate. In particular, cancellation for a single pair of rows would not by itself control the largest eigenvalue when the number of rows grows polynomially with X . The required statement is a weighted matrix estimate for the complete family of growing slopes $m \asymp Q$.

7.3 Affine gcd separation

The two affine targets in (144) have a simple exact resultant.

Lemma 7.3 (Gcd and very-high-prime separation). *Let $m \neq m'$ and suppose both targets at the same physical time j are active very-high values:*

$$n = mj + h_0 = pr, \quad n' = m'j + h_0 = p'r', \quad p, p' > y. \quad (145)$$

Then

$$\boxed{\gcd(n, n') \mid h_0(m - m').} \quad (146)$$

Moreover

$$p \neq p', \quad p \nmid n', \quad p' \nmid n, \quad (147)$$

and hence neither very-high prime divides the other target's cofactor.

Proof. Every common divisor of n and n' divides the integer linear combination

$$m'n - mn' = h_0(m' - m),$$

which proves (146). If $p \mid n'$, then, because $p \mid n$, one has $p \mid (m - m')j$. The inequality $p > J_+$ excludes $p \mid j$, and $|m - m'| \leq \Delta_M < p$ excludes $p \mid m - m'$. Thus $p \nmid n'$. Interchanging the two rows gives $p' \nmid n$, and in particular $p \neq p'$. Since $r' \mid n'$ and $r \mid n$, the final assertion follows. $\square$

The gcd restriction controls local common factors, but it does not assign a sign to $\mu(n)\mu(n')$. It therefore cannot be entered as a cancellation estimate for $\Xi_{\neq p}$.

7.4 Determinant reindexing

For fixed distinct rows m, m' , let $\mathcal{S}_{m, m'}$ consist of quadruples (p, r, p', r') satisfying all of the following:

(i) $p, p' > y$ are primes and $pr, p'r'$ lie on the declared squarefree high support;

(ii)

$$m'pr - mp'r' = h_0(m' - m); \quad (148)$$

(iii) the common integer

$$j = \frac{pr - h_0}{m} = \frac{p'r' - h_0}{m'} \quad (149)$$

belongs to the fixed physical orbit slice; and

(iv) $p = P^+(pr)$, $p' = P^+(p'r')$, and both recovered cofactors satisfy the literal missing tests.

All inherited residue, profile, and endpoint conditions remain in this definition; the determinant equation does not enlarge the carrier.

Proposition 7.4 (Exact determinant form of one parity entry). *The map*

$$j \mapsto (p_m(j), r_m(j), p_{m'}(j), r_{m'}(j)) \quad (150)$$

is a bijection from the jointly missing times in (144) onto $\mathcal{S}_{m,m'}$. Consequently

$$\boxed{(U_d \Xi_{\neq p} U_d)_{m,m'} = \frac{1}{\sqrt{W_M(m)W_M(m')}} \sum_{(p,r,p',r') \in \mathcal{S}_{m,m'}} \mu(r)\mu(r') \times \overline{a_m(j)} a_{m'}(j) \Gamma_{m,m'}(j),} \quad (151)$$

where j is given by (149).

Proof. For a jointly missing time, unique largest-prime factorization supplies (p, r) and (p', r') . Multiplying the first equality in (145) by m' , the second by m , and subtracting gives (148). Thus the displayed map lands in $\mathcal{S}_{m,m'}$, and it is injective because either product pr recovers j .

Conversely, conditions (ii)–(iii) imply both affine equalities at the same physical time, while (i) and (iv) give exactly the active high and missing tests. Hence every quadruple comes from one jointly missing time. Finally, squarefreeness gives

$$\mu(pr)\mu(p'r') = (-\mu(r))(-\mu(r')) = \mu(r)\mu(r'),$$

so (144) becomes (151). $\square$

This reformulation exposes a possible bilinear or dispersion route, but it does not prove that the signed determinant sum is smaller than its absolute majorant. The primes p, p' are distinct, yet distinctness alone gives no Möbius cancellation in r, r' .

7.5 Why the opposite-row mask does not diagonalize the kernel

For a fixed time j , put

$$B(j) := \sum_{\gamma} |q_{\gamma}(j)|^2 \quad (152)$$

and, when $B(j) > 0$, define

$$\eta_m(j) := \frac{1}{B(j)} \sum_{\gamma: K_{m\gamma}=0} |q_{\gamma}(j)|^2. \quad (153)$$

Set $\eta_m(j) = 0$ when $B(j) = 0$.

Proposition 7.5 (Exact two-row mask overlap). *For every m, m', j ,*

$$\boxed{(1 - \eta_m(j) - \eta_{m'}(j))_+ B(j) \leq \Gamma_{m,m'}(j) \leq B(j).} \quad (154)$$

On a nondegenerate complete direct opposite-row cell to which the TPC–50 weighted mask theorem applies uniformly,

$$\sup_{m,j} \eta_m(j) = o(1) \implies \Gamma_{m,m'}(j) = (1 - o(1))B(j) \quad (155)$$

uniformly for all active m, m', j .

Proof. The product $K_{m\gamma}K_{m'\gamma}$ vanishes only on the union of the two forbidden opposite-row sets. The weight of that union is at most the sum of their weights, which gives the lower bound. The upper bound is immediate from $0 \leq K_{m\gamma}K_{m'\gamma} \leq 1$. The final statement follows from the uniform one-row defect estimate of TPC-50 and the same union bound [3]. $\square$

Thus static-mask survival has zero positive-power cost on the declared interior cell, but it points in the opposite direction from an orthogonality argument: two source rows retain almost the same opposite-row mass. The mask does not supply the open spectral estimate for $\Xi_{\neq p}$. Any such estimate must use the signed affine correlation in (144), the signed determinant form in (151), or additional literal structure not used here.

7.6 The exact open spectral target

The preceding results identify, but do not estimate, the remaining uniform quantity. Assume $S_M \neq \{0\}$; when $S_M = \{0\}$, the missing map vanishes and this gate is vacuous. Put

$$\rho_{\text{par}}(X) := \lambda_{\max}(\Xi_{\neq p}) = \lambda_{\max}(U_d \Xi_{\neq p} U_d) \geq 0. \quad (156)$$

The nonnegativity follows because $\Xi_{\neq p}$ is Hermitian with zero trace. By (114),

$$\boxed{\kappa_M = 1 + \rho_{\text{par}}(X)}. \quad (157)$$

For one distinguished vector, only the corresponding Rayleigh quotient in (132) is needed. A bound for that one quadratic form is strictly weaker than (156) and must remain labeled distinguished.

Neither the gcd lemma, the determinant bijection, Brun–Titchmarsh on one cofactor ladder, nor the static-mask overlap proves $\rho_{\text{par}}(X) = X^{o(1)}$. Treating $\mu(mj + h_0)$ as an independent random sign would add an unproved hypothesis and is not part of the TPC interface.

8 Weighted collision certificates and sharp tests

The constant κ_M is an exact spectral target, but a direct eigenvalue computation is not always the best proof. We now express it as the Gram of a literal weighted incidence matrix and give several rigorous certificates: collision-graph Schur tests, stable-rank obstructions, exact Schur complements, and blockwise estimates. These statements retain the fixed physical coefficients. Replacing their signed entries by support cardinalities is explicitly identified as a sufficient relaxation.

8.1 The normalized physical incidence matrix

Let

$$F_M := M_X \mathbf{D}_M^{\dagger/2}|_{S_M} : S_M \longrightarrow \ell^2(\mathcal{I}_{\text{out}}) \quad (158)$$

and write its matrix as

$$f_{i,m} = \frac{1}{\sqrt{W_M(m)}} \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i, m(w)=m}} c_w \quad (W_M(m) > 0). \quad (159)$$

The fixed- (i, m) decoding discussed after theorem 6.4 makes the sum in (159) either empty or a single literal atom. Define the row-collision Gram

$$\mathbf{K}_M := F_M^* F_M. \quad (160)$$

Then

$$(\mathbf{K}_M)_{m,n} = \sum_i \overline{f_{i,m}} f_{i,n}, \quad \kappa_M = \lambda_{\max}(\mathbf{K}_M). \quad (161)$$

Every off-diagonal summand represents two different source rows landing at the same native output. By the prime-block singleton, their terminal primes are unequal. According to whether their residual labels agree, the same summand belongs to $\Xi_{=s}$ or $\Xi_{\neq s}$.

Theorem 8.1 (Unit columns, stable rank, and dimension obstruction). *Let*

$$r_M := \dim S_M, \quad d_M^{\text{out}} := \text{rank } F_M. \quad (162)$$

If $r_M > 0$, then every column of F_M has norm one and

$$(\mathbf{K}_M)_{m,m} = 1 \quad (m \in \mathcal{M}_M), \quad (163)$$

$$\|F_M\|_{\text{HS}}^2 = \text{tr } \mathbf{K}_M = r_M, \quad (164)$$

$$\boxed{\kappa_M = \frac{r_M}{\text{srank}(F_M)} \geq \frac{r_M}{d_M^{\text{out}}},} \quad (165)$$

$$\boxed{1 \leq \kappa_M \leq \min\{d_{\text{miss}}, r_M\}.} \quad (166)$$

Moreover:

- (i) $\kappa_M = 1$ if and only if the normalized physical row images are orthonormal;
- (ii) $\kappa_M = r_M$ if and only if F_M has rank one.

Proof. Fix an active source row m . Distinct atoms driven by m cannot occupy the same native coordinate: i fixes j , and m, j, h_0 fix n , its unique very-high prime, its cofactor, and its residual label. Therefore

$$\sum_i |f_{i,m}|^2 = \frac{1}{W_M(m)} \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |c_w|^2 = 1.$$

This proves (163) and (164). By definition,

$$\text{srank}(F_M) = \frac{\|F_M\|_{\text{HS}}^2}{\|F_M\|^2} = \frac{r_M}{\kappa_M},$$

and $\text{srank}(F_M) \leq \text{rank } F_M$, proving (165). The lower bound one follows from the trace, the upper bound r_M follows because the largest eigenvalue of a positive matrix is at most its trace, and the d_{miss} -bound is [corollary 6.3](#). If the largest eigenvalue is one, the trace forces all r_M eigenvalues to equal one, hence $\mathbf{K}_M = I$. If it equals r_M , every other eigenvalue is zero, which is equivalent to rank one. $\square$

Corollary 8.2 (Asymptotic rank kill for a subpower degree). *Suppose*

$$r_M = X^{\rho_M + o(1)}, \quad d_M^{\text{out}} \leq X^{\omega_M + o(1)}. \quad (167)$$

Then

$$\boxed{\kappa_M \geq X^{(\rho_M - \omega_M)_+ + o(1)}}. \quad (168)$$

In particular, a proof of $\kappa_M = X^{o(1)}$ requires the actual native image rank to be within a subpower factor of the number of active missing rows.

This is a kill test for the *mass-degree certificate*, not for the full reassembly problem. A distinguished coefficient vector can avoid the large-singular-value direction even when (168) is polynomial.

8.2 The weighted collision graph

Let $\mathcal{G}_M$ be the graph on active source rows with

$$m \sim n \iff m \neq n \text{ and } (\mathbf{K}_M)_{m,n} \neq 0, \quad (169)$$

and attach the signed edge weight $(\mathbf{K}_M)_{m,n}$. A larger support graph may instead join two rows whenever they share some output; the signed graph is obtained after all literal phases on the shared outputs have been summed.

Theorem 8.3 (Component reduction and weighted Schur certificate). *Let $\mathcal{C}$ be the connected components of $\mathcal{G}_M$. Then*

$$\kappa_M = \max_{C \in \mathcal{C}} \lambda_{\max}((\mathbf{K}_M)_{C,C}). \quad (170)$$

For every family of positive row weights $t_m > 0$,

$$\kappa_M \leq \max_m \left\{ 1 + \sum_{n \neq m} |(\mathbf{K}_M)_{m,n}| \frac{t_n}{t_m} \right\}. \quad (171)$$

In particular, if

$$\Delta_{\text{coh}} := \max_m \sum_{n \neq m} |(\mathbf{K}_M)_{m,n}|, \quad (172)$$

then

$$\kappa_M \leq 1 + \Delta_{\text{coh}}. \quad (173)$$

Proof. There is no matrix entry between different connected components, so $\mathbf{K}_M$ is their orthogonal direct sum, proving (170). Entrywise absolute value and the Schur test give

$$\|\mathbf{K}_M\| \leq \|\mathbf{K}_M\| \leq \max_m \frac{1}{t_m} \sum_n |(\mathbf{K}_M)_{m,n}| t_n.$$

Use the unit diagonal from (163). Taking $t_m = 1$ gives the last claim. $\square$

Thus a subpower bound for the *weighted coherent degree*, rather than the raw number of atoms in one output fiber, is sufficient for $\kappa_M = X^{o(1)}$. The weights t_m permit exceptional rows with small physical mass or asymmetric overlap to be handled without replacing the physical atomic norm.

8.3 A phase-blind overlap certificate

For an active row m , put

$$I_m := \{i : f_{i,m} \neq 0\}. \quad (174)$$

For two distinct active rows define the directed weighted overlap

$$\theta_{m \rightarrow n} := \sum_{i \in I_m \cap I_n} |f_{i,m}|^2. \quad (175)$$

By the unit-column identity, $0 \leq \theta_{m \rightarrow n} \leq 1$.

Proposition 8.4 (Weighted-codegree bound). *For distinct active rows,*

$$|(\mathbf{K}_M)_{m,n}| \leq \sqrt{\theta_{m \rightarrow n} \theta_{n \rightarrow m}}. \quad (176)$$

Consequently

$$\kappa_M \leq 1 + \max_m \sum_{n \neq m} \sqrt{\theta_{m \rightarrow n} \theta_{n \rightarrow m}}. \quad (177)$$

Proof. Only the common outputs $I_m \cap I_n$ occur in $(\mathbf{K}_M)_{m,n}$. Cauchy–Schwarz on that set gives (176); insert it in (173). $\square$

The overlap certificate is deliberately phase blind. Its failure does not prove that κ_M is large: the exact signed sum in $(\mathbf{K}_M)_{m,n}$ may be much smaller than its Cauchy–Schwarz envelope.

For completeness, the incidence matrix itself admits the following rectangular Schur test. If positive weights a_i, b_m and constants A, B satisfy

$$\sum_m |f_{i,m}| b_m \leq A a_i, \quad \sum_i |f_{i,m}| a_i \leq B b_m, \quad (178)$$

then

$$\boxed{\kappa_M = \|F_M\|^2 \leq AB.} \quad (179)$$

This form can be more convenient when row and output weights factor into literal smooth profiles.

8.4 Exact Schur-complement certification

The preceding Schur bounds use absolute values. The following criterion retains all signed cancellations and is exact. Split the active row space as $S_M = S_U \oplus S_V$ and write

$$\mathbf{K}_M = \begin{pmatrix} K_{UU} & K_{UV} \\ K_{VU} & K_{VV} \end{pmatrix}. \quad (180)$$

Proposition 8.5 (Exact recursive upper certificate). *If $\Lambda > \lambda_{\max}(K_{VV})$, then*

$$\lambda_{\max}(\mathbf{K}_M) \leq \Lambda \quad (181)$$

if and only if

$$\boxed{K_{UU} + K_{UV}(\Lambda I - K_{VV})^{-1} K_{VU} \preceq \Lambda I.} \quad (182)$$

At the boundary where $C := \Lambda I - K_{VV}$ may be singular, the exact conditions are

$$C \succeq 0, \quad (183)$$

$$\text{Ran } K_{VU} \subseteq \text{Ran } C, \quad (184)$$

$$\Lambda I - K_{UU} - K_{UV} C^\dagger K_{VU} \succeq 0. \quad (185)$$

Proof. The target is equivalent to positivity of

$$\Lambda I - \mathbf{K}_M = \begin{pmatrix} \Lambda I - K_{UU} & -K_{UV} \\ -K_{VU} & \Lambda I - K_{VV} \end{pmatrix}.$$

The ordinary Schur complement gives (182) when the lower-right block is positive definite. The Moore–Penrose positivity criterion for a Hermitian block matrix gives (183)–(185). $\square$

This certificate permits a dense exceptional core V to be eliminated exactly while a large sparse part U is treated recursively. A floating-point positive gap at finitely many values of X is not such a certificate; exact arithmetic or interval enclosures must also control its scale dependence.

8.5 Blockwise physical certificates

Let the missing carrier have a fixed disjoint partition

$$\mathcal{W}_M = \bigsqcup_{b=1}^B \mathcal{W}_{M,b}, \quad M_X = \sum_{b=1}^B M_b, \quad \mathbf{D}_M = \sum_{b=1}^B \mathbf{D}_b, \quad (186)$$

where $\mathbf{D}_b$ is the raw diagonal of the b -th block. Put

$$S_b := \text{Ran } \mathbf{D}_b, \quad F_b := M_b \mathbf{D}_b^{\dagger/2}|_{S_b}, \quad \kappa_b := \|F_b\|^2. \quad (187)$$

Theorem 8.6 (Block reassembly upper bounds). *For every fixed carrier partition,*

$$\boxed{\kappa_M \leq \sum_{b=1}^B \kappa_b.} \quad (188)$$

More precisely, define

$$a_{bc} := \|F_b^* F_c\|, \quad (189)$$

where the maps are extended by zero between their active row spaces. Then

$$\boxed{\kappa_M \leq \left\| [a_{bc}]_{1 \leq b, c \leq B} \right\|_{\ell_B^2 \rightarrow \ell_B^2}.} \quad (190)$$

In particular:

- (i) if $\text{Ran } F_b \perp \text{Ran } F_c$ for $b \neq c$, then $\kappa_M \leq \max_b \kappa_b$;
- (ii) without any cross-block orthogonality, $a_{bc} \leq \sqrt{\kappa_b \kappa_c}$, which recovers (188).

Proof. For ζ , set $x_b = \mathbf{D}_b^{1/2} \zeta \in S_b$. Then

$$M_X \zeta = \sum_b F_b x_b, \quad \sum_b \|x_b\|^2 = \mathcal{D}_M(\zeta).$$

The triangle inequality followed by Cauchy–Schwarz gives

$$\left\| \sum_b F_b x_b \right\|^2 \leq \left(\sum_b \sqrt{\kappa_b} \|x_b\| \right)^2 \leq \left(\sum_b \kappa_b \right) \left(\sum_b \|x_b\|^2 \right),$$

which proves (188). On the orthogonal direct sum $\bigoplus_b S_b$, the Gram of the row operator $(x_b)_b \mapsto \sum_b F_b x_b$ has blocks $F_b^* F_c$. The norm of a block operator matrix is at most the norm of the scalar matrix of block norms, proving (190). The two special cases follow immediately. $\square$

If the carrier is divided into $B = O(\log X)$ dyadic cofactor blocks and one proves

$$\kappa_b = X^{o(1)} \quad \text{uniformly in } b, \quad (191)$$

then (188) gives

$$\boxed{\kappa_M = X^{o(1)}} \quad (192)$$

without any cross-block orthogonality hypothesis. What is not automatic is (191): on a compatible endpoint block, the ambient determinant-ladder bound still contains the positive-power factor $1 + L/R$.

8.6 Sharp examples

Example 8.7 (Complete coherent incidence). *Take r source rows and one native output, with one atom of coefficient one in every row. Then*

$$\mathbf{D}_M = I_r, \quad F_M = [1 \ \cdots \ 1], \quad d_{\text{miss}} = r, \quad \kappa_M = r. \quad (193)$$

Thus the ambient degree bound can be exactly sharp.

Example 8.8 (Same support and magnitudes, Fourier cancellation). *Take r rows and r native outputs, with one atom at every pair (i, m) , all of modulus $r^{-1/2}$. The row weights are one and every native fiber has degree r . If all phases are equal, then*

$$F_M = r^{-1/2} \mathbf{1} \mathbf{1}^\top, \quad \kappa_M = r.$$

For the Fourier phases

$$(F_M)_{i,m} = r^{-1/2} \exp(2\pi i im/r), \quad (194)$$

F_M is unitary and

$$\boxed{d_{\text{miss}} = r, \quad \kappa_M = 1.} \quad (195)$$

The two arrays have identical support, identical atomic magnitudes, and identical native degrees. Hence support and raw mass do not determine the restricted native norm. This is an abstract sharpness example; it does not assert that the literal arithmetic phases form a Fourier matrix.

Example 8.9 (Uniform and distinguished scopes separate sharply). *For*

$$F_M = [1 \ 1]$$

one has $\kappa_M = 2$. The distinguished normalized coefficient $x = (1, -1)$ has pointwise ratio zero, whereas $x = (1, 1)$ attains the uniform value two. A good distinguished ratio therefore gives no coefficient-uniform bound, while a large uniform norm does not prove that the inherited physical vector occupies its top singular direction.

Example 8.10 (Either stage of the residual quotient can carry the loss). *If T_M embeds r coefficient coordinates as the r residual labels over one native output, then $\kappa_{\text{res}} = 1$, $\beta_M = r$, and $\kappa_M = r$. Conversely, if T_M coherently groups r row coordinates into one residual coordinate and $\mathbf{E}$ is the identity on that coordinate, then $\kappa_{\text{res}} = r$, $\beta_M = 1$, and again $\kappa_M = r$. Thus neither factor in (127) can be omitted.*

8.7 Exponent consequence for the reassembly ledger

Suppose, in a stated uniform or distinguished scope,

$$\kappa_M \leq X^{\nu_M + o(1)}, \quad (196)$$

$$\mathcal{D}_M(\zeta) \leq X^{-\sigma_M + o(1)} \mathcal{D}_{\text{ref}}(\zeta), \quad (197)$$

$$\|A_X \zeta\|^2 \geq X^{-\lambda_{\text{rep}} + o(1)} \mathcal{D}_{\text{ref}}(\zeta). \quad (198)$$

Then

$$q_X(\zeta) := \frac{\kappa_M \mathcal{D}_M(\zeta)}{\|A_X \zeta\|^2} \leq X^{\nu_M + \lambda_{\text{rep}} - \sigma_M + o(1)}. \quad (199)$$

Consequently

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}} \implies \frac{\|(A_X + M_X)\zeta\|^2}{\|A_X \zeta\|^2} = 1 + o(1).} \quad (200)$$

The upper half of the last equality follows from the triangle inequality, and the lower half from (133).

At exponent equality one must retain the actual margin. If

$$1 - \sqrt{q_X(\zeta)} \geq X^{-a+o(1)} \quad (201)$$

for some $a \geq 0$, then the certificate charges at most

$$\lambda_{\text{reasm}} \leq 2a \quad (202)$$

in the matching scope. This cost must be entered in the strict $1/400$ ledger. If q_X is not bounded below one, the present reverse-triangle certificate is silent; that does not prove failure of a signed or shorted-Gram route.

The dimension obstruction (168) gives the necessary lower scale

$$\nu_M \geq (\rho_M - \omega_M)_+.$$

Thus a zero-cost mass-degree proof cannot follow from the available savings unless

$$\sigma_M > (\rho_M - \omega_M)_+ + \lambda_{\text{rep}}. \quad (203)$$

Failure of (203) closes this particular uniform certificate, not the distinguished signed route.

At the critical cutoff, the coefficient-blind residual capacity exponent from TPC-60 is at least

$$\frac{15077}{105000} \approx 0.143590,$$

far above $1/400$. The new quantity ν_M is not that ambient capacity exponent. A proof of $\Delta_{\text{coh}} = X^{o(1)}$, of the weighted overlap condition in (177), or of the block condition (191) would instead give $\nu_M = 0$. None of those arithmetic estimates is asserted here.

9 Critical-cutoff ledger and route decisions

The exposure and Gram identities are exact at every finite X . Their role in the endpoint argument is determined only after missing mass, restricted grouping, and the represented reference have been measured on the same carrier and with the same quantifier. This section performs that comparison and records which numerical differences are genuine tests rather than additive losses.

9.1 The critical arithmetic wedge

At the smallest very-high cutoff allowed by the fixed support constants, put

$$\theta_J := \frac{133}{400}, \quad \lambda_L := \frac{99979}{210000}. \quad (204)$$

Thus $J = X^{\theta_J+o(1)}$, and a critical cofactor block has $R = X^{\theta_J+o(1)}$. The coefficient-blind residual determinant-ladder degree on that block has exponent

$$\mu_{\text{res}}^{\text{amb}} := \lambda_L - \theta_J = \frac{15077}{105000} = 0.143590 \dots \quad (205)$$

This number is much larger than $1/400$, but it is not by itself an endpoint loss. In the mass-degree certificate it is multiplied by the missing mass before the reverse triangle inequality is applied.

Suppose the missing set in the critical block has cardinality

$$|E_m(R)| \leq X^{e+o(1)} \quad (206)$$

uniformly in the stated row class, and suppose the balanced-exposure hypothesis $H_m^{\text{eff}} \geq X^{\theta_J + o(1)}$ holds there. Then [corollary 5.2](#) gives the conditional missing-mass saving

$$\sigma_{\text{cof}} = \theta_J - e. \quad (207)$$

Consequently the residual degree and cofactor saving leave the algebraic window

$$\boxed{2\theta_J - \lambda_L = \frac{39671}{210000} = 0.188909\dots} \quad (208)$$

Let the actual residual-label erasure on this block satisfy

$$\beta_{M,R} \leq X^{\mu_{\text{erase}} + o(1)}, \quad (209)$$

and let the represented map obey

$$\|A_X \zeta\|^2 \geq X^{-\lambda_{\text{rep}} + o(1)} \mathcal{D}_H(\zeta) \quad (210)$$

in the same uniform or distinguished scope. Paying the ambient residual degree, but only the restricted erasure norm, gives

$$\frac{\|M_{X,R} \zeta\|^2}{\|A_X \zeta\|^2} \leq X^{e + \mu_{\text{erase}} + \lambda_{\text{rep}} - (2\theta_J - \lambda_L) + o(1)}. \quad (211)$$

Hence this certificate has zero fixed-power reassembly cost under the strict condition

$$\boxed{e + \mu_{\text{erase}} + \lambda_{\text{rep}} < \frac{39671}{210000}.} \quad (212)$$

For $O(\log X)$ compatible blocks, the same conclusion follows only when the corresponding exponent inequality holds block by block with one fixed positive margin $\delta > 0$. Then each block norm is $O(X^{-\delta/2 + o(1)} \|A_X \zeta\|)$, and the triangle inequality over the logarithmic family remains $o(\|A_X \zeta\|)$. Mere pointwise strictness with a margin tending to zero is not a uniform asymptotic certificate. None of the three hypotheses in (206)–(210) is proved by the numerical window itself.

9.2 Why the direct ambient door remains closed

The inherited coefficient-blind native degree has exponent $267/400$. Even the smallest nonempty missing set at the critical cofactor scale has $e = 0$, so the balanced harmonic certificate saves at most $\theta_J = 133/400$. A direct phase-blind payment therefore leaves

$$\boxed{\frac{267}{400} - \frac{133}{400} = \frac{67}{200} > 0.} \quad (213)$$

This proves that the particular proof architecture which combines the best pure-cardinality missing saving with the full ambient native degree cannot make the perturbation $o(1)$. It does not lower-bound κ_M , does not rule out cancellation in the parity kernel, and is not a loss to be entered into the final ledger.

The residual window in (208) and the direct gap in (213) therefore have different logical meanings. The first is conditional room for a two-stage proof; the second closes one support-only proof. Neither is an arithmetic lower bound for the full physical map.

9.3 The optimal one-stage effective-product test

The restricted Gram eliminates the need to commit in advance to the residual factorization. Fix a reference diagonal $\mathcal{D}_{\text{ref}}$ and suppose, in one declared scope,

$$\mathcal{D}_M(\zeta) \leq X^{-\sigma_M+o(1)}\mathcal{D}_{\text{ref}}(\zeta), \quad (214)$$

$$\kappa_M \leq X^{\nu_M+o(1)}, \quad (215)$$

$$\|A_X\zeta\|^2 \geq X^{-\lambda_{\text{rep}}+o(1)}\mathcal{D}_{\text{ref}}(\zeta). \quad (216)$$

Then the exact perturbation parameter is bounded by

$$q_X(\zeta) := \frac{\kappa_M \mathcal{D}_M(\zeta)}{\|A_X\zeta\|^2} \leq X^{\nu_M+\lambda_{\text{rep}}-\sigma_M+o(1)}. \quad (217)$$

Thus

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}} \implies \lambda_{\text{reasm}} = 0} \quad (218)$$

for a coefficient-uniform statement, and the analogous conclusion holds with $\kappa_M(\zeta_*)$ and $\lambda_{\text{reasm},*} = 0$ for one distinguished physical vector. These scopes cannot be interchanged.

At equality of exponents, one must use the actual quantity q_X . If

$$1 - \sqrt{q_X(\zeta)} \geq X^{-a+o(1)}, \quad (219)$$

then the reverse-triangle certificate charges at most

$$\lambda_{\text{reasm}} \leq 2a \quad (220)$$

in the matching scope. If $q_X \geq 1$, that certificate is silent. It does not imply destructive interference, because the exact cross term may still be favorable and the TPC-60 shorted-Gram route remains available.

The rank obstruction is a useful early test. If $r_M = X^{\rho_M+o(1)}$ and $\text{rank } F_M \leq X^{\omega_M+o(1)}$, then

$$\nu_M \geq (\rho_M - \omega_M)_+. \quad (221)$$

Accordingly, the uniform mass-degree route cannot satisfy (218) unless

$$\sigma_M > (\rho_M - \omega_M)_+ + \lambda_{\text{rep}}. \quad (222)$$

Failure closes this certificate but need not close a distinguished vector orthogonal to the top singular directions.

9.4 The complete endpoint ledger

The prospective fixed-shift transfer chain remains the one inherited from TPC-60. Its losses are recorded as

$$\boxed{\begin{aligned} \Lambda_{\text{tot}} := & \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} \\ & + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}. \end{aligned}} \quad (223)$$

The endpoint requirement is

$$\boxed{\Lambda_{\text{tot}} < \frac{1}{400}}. \quad (224)$$

For a distinguished chain, λ_{reasm} is consistently replaced by $\lambda_{\text{reasm},*}$. Equality is rejected unless the leading constants and all remainders are controlled strongly enough to recover a strict final inequality.

In particular, in the equality-scale situation (219)–(220), put $\Lambda_{\text{other}} := \Lambda_{\text{tot}} - \lambda_{\text{reasm}}$. The reverse-triangle certificate can fit the endpoint only if

$$\boxed{2a + \Lambda_{\text{other}} < \frac{1}{400}.} \quad (225)$$

Thus the number a is not free slack: it consumes twice its value in the loss ledger.

The quantities e , σ_M , ν_M , μ_{erase} , and $67/200$ are not additional arrows in (223). They are inputs or diagnostics used to decide the single reassembly loss. When λ_{rep} is assembled from λ_H , λ_{aff} , or other arrows already listed in that ledger, it enters the comparison tests above but is not charged a second time. Likewise, the cutoff $y \asymp Q$, exact coordinate changes, Moore–Penrose restriction to the active support, and finite or logarithmic block decompositions have zero fixed-power cost.

9.5 Stop and pivot rules

The current proof architecture has the following exact decisions.

1. If some active row has $W_R(m) = 0 < W_M(m)$, the coefficient-uniform represented-mass route is dead. An average over other rows cannot repair it.
2. If the dimension exponents force

$$(\rho_M - \omega_M)_+ + \lambda_{\text{rep}} \geq \sigma_M,$$

then the uniform restricted mass–degree certificate cannot satisfy (218) at the proposed exponents. The distinguished and signed routes remain separate questions.

3. If one pays the complete ambient native degree, the positive gap (213) closes that proof even before the other endpoint losses are charged.
4. If a literal coefficient vector satisfies $A_X \zeta \neq 0$ but $(A_X + M_X) \zeta = 0$, the TPC–60 uniform shorted-Gram gate is exactly dead.
5. If a proved necessary loss reaches $1/400$, or the sum in (223) reaches it without a quantified constant-level margin, the endpoint route must stop.

None of these failures may be repaired by averaging over shifts, replacing the fixed h_0 , renormalizing the raw atoms by fiber sizes, deleting coefficient-dependent zeros, or substituting a pointwise Rayleigh quotient for a uniform operator norm. Those operations change the mathematical problem rather than close the stated gate.

10 Result ledger, claim boundary, and the next gate

TPC–61 replaces two coarse quantities from TPC–60 by literal arithmetic objects. The missing mass is the realized cofactor measure outside the fixed retained relation, and the effective native degree is the largest eigenvalue of the normalized physical row Gram. The latter has now been calibrated to a two-affine-form Möbius kernel. These are reductions and certificates; the required exposure and spectral estimates remain open.

10.1 Result ledger

Object	Exact conclusion	Status
Literal high carrier	One fixed h_0 , the raw coefficient $b_w = \beta_w \zeta_{m(w)}$, and the canonical decoding $n = mj + h_0 = pr$, $p = P^+(n) > y$, $s = d_m r$.	L1 interface
Realized cofactor measure	The row measure $\nu_m(r)$ gives exact identities $W_R(m) = \nu_m(\mathcal{I}_m)$ and $W_M(m) = \nu_m(\mathcal{I}_m^c)$, followed by the sharp rowwise minimax.	L1 identity and L0 minimax
Opaque-carrier test	The inherited finite-relation hypotheses alone permit a missing-only active row, so they imply neither positive uniform exposure nor a fixed-power missing-mass saving.	Interface no-go
Cofactor ladder	For fixed m, r , all solutions are $j = j_0 + rt$, $p = p_0 + mt$; Brun–Titchmarsh gives the literal harmonic upper capacity.	L1 arithmetic upper bound
Harmonic defect	$\varepsilon_m \ll \mathfrak{h}_m(E_m)/H_m^{\text{eff}}$, with the exact dyadic exponent law. Its usefulness requires a small retained-relation complement and a lower effective-exposure bound.	L1 conditional certificate
Residual/native audit	A positive critical residual wedge remains, whereas paying the full direct native degree has a fixed exponent deficit 67/200.	Route-specific opening and closure
Restricted native degree	κ_M is the optimal generalized-Gram eigenvalue on the actual row-coefficient image, with $1 \leq \kappa_M \leq \min\{d_{\text{miss}}, \dim S_M\}$ when the carrier is nonempty.	L0 theorem; L1 specialization
Prime/residual alias split	On a nonempty missing carrier, same-prime blocks are isometric and $\kappa_M = 1 + \lambda_{\max}(\Xi_{\neq p})$; the latter splits into same- and unequal-residual alias operators.	L0/L1 exact Gram
Collision certificates	Stable rank, weighted Schur, codegree, exact Schur-complement, and block certificates give falsifiable upper or lower tests for κ_M .	L0 theorems
Parity calibration	A fixed row gauge turns every off-diagonal alias into the weighted kernel $\mu(mj + h_0)\mu(m'j + h_0)$; equivalently it is a signed sum on $m'pr - mp'r' = h_0(m' - m)$.	L1 identity
Static-mask overlap	On the complete nondegenerate direct cell, two rows retain $1 - o(1)$ of the same opposite-row mass. The mask costs no positive power but does not diagonalize the parity kernel.	L0 inequality and L1 consequence
Reassembly	If $\sigma_M > \nu_M + \lambda_{\text{rep}}$, the effective mass-degree certificate gives zero fixed-power reassembly loss in the matching uniform or distinguished scope.	Conditional only

10.2 Six-item physical audit

The statements above pass the interface audit in the following precise sense. Passing this audit verifies that the right physical object is being studied; it does not prove a favorable exposure or spectral estimate.

- (A1) *Literal coefficients.* Every atom has the single inherited coefficient $\beta_w \zeta_{m(w)}$. The signs $-\mu(s(w))$ in M_X are the physical high-character signs. The row gauge U_d is a fixed

unitary change of coordinates, not a choice of favorable atom phases.

- (A2) *Fixed physical shift.* Every occurrence of a target, cofactor ladder, gcd identity, and determinant equation comes from $n = mj + h_0$ for the same fixed nonzero h_0 . No shift average or frozen family is substituted.
- (A3) *Physical atomic normalization.* The measures ν_m , the weights $W_M(m)$, and the matrix $\mathbf{D}_M$ use raw counting-measure atoms. No division by a native fiber size is hidden in the input norm.
- (A4) *Canonical active representation.* The row $m = d_m \ell_m$, largest prime $p = P^+(n)$, cofactor $r = n/p$, residual label $s = d_m r$, and native output i are canonically recovered. Moore–Penrose normalization is restricted to $S_M = \text{Ran } \mathbf{D}_M$; only identically inactive rows disappear.
- (A5) *Actual support.* The carriers $\mathcal{W}_H, \mathcal{W}_R, \mathcal{W}_M$ and the relation $\mathcal{I}_X$ are fixed before ζ is chosen. The identity $w \in \mathcal{W}_M \iff r(w) \notin \mathcal{I}_{m(w)}$ is used only under the explicitly declared identical-support specialization.
- (A6) *Endpoint ledger.* Exact disintegration, gauge calibration, determinant reindexing, and finite/logarithmic partitions carry no fixed-power loss. A proved critical-margin loss is charged only as λ_{reasm} , and every other arrow remains in the strict $1/400$ sum.

10.3 What is not proved

The following conclusions do not follow from this paper.

1. The actual upstream relation $\mathcal{I}_X$ has not been shown to meet $\text{supp } \nu_m$ on every active row, nor has its harmonic complement been shown small.
2. No coefficient-uniform or distinguished fixed-power bound $\mathcal{D}_M \leq X^{-\sigma_M + o(1)} \mathcal{D}_{\text{ref}}$ is proved.
3. No lower bound for the represented reference $\|A_X \zeta\|^2$ is supplied; critical-window occupancy and terminal activation remain separate upstream gates.
4. The exact parity target $\rho_{\text{par}}(X) = \lambda_{\text{max}}(\Xi_{\neq p})$ is not shown to be $X^{o(1)}$, or to satisfy any smaller fixed exponent than the available missing-mass saving.
5. The gcd divisibility $\gcd(mj + h_0, m'j + h_0) \mid h_0(m - m')$, the distinctness of the two high primes, and the determinant parametrization do not imply cancellation of the Möbius product.
6. The $1 - o(1)$ static-mask overlap is not an orthogonality theorem. In fact it shows that the opposite-row coordinate alone supplies no polynomial separation of two source rows.
7. No bound for one row pair is promoted to an operator-norm bound over the polynomially growing row family.
8. No positive physical shorted-Gram lower bound, no value $\lambda_{\text{reasm}} = 0$, no fixed-shift parity improvement, no prime-pair lower bound, no prime-pair asymptotic, and no twin-prime implication is proved.

In particular, treating $\mu(mj + h_0)$ as an independent random sign would be a new unproved hypothesis. Likewise, a result for fixed coefficients, a logarithmically averaged correlation, or an average over shifts cannot be inserted into (144) without verifying the actual growing-slope, physical-weight, and operator-norm quantifiers.

10.4 The ordered next gate

The next work should preserve the following order. A favorable spectral model cannot repair an empty represented carrier, while a favorable distinguished Rayleigh quotient cannot be entered as a uniform constant.

Door A. *Recover the actual retained relation.* Reopen the provenance of $\mathcal{I}_X$ and evaluate the literal row intersections

$$W_R(m) = \nu_m(\mathcal{I}_m), \quad W_M(m) = \nu_m(\mathcal{I}_m^c). \quad (226)$$

The uniform route is killed immediately by an active row with $W_R(m) = 0 < W_M(m)$. Equal row multiplicities, equal dyadic capacities, or positive exposure on other rows do not repair it. A distinguished route must instead retain the weights $W_H(m)|z_m|^2$ exactly.

Door B. *Prove a missing-mass saving in one declared scope.* A uniform target has the form

$$\mathcal{D}_M(\zeta) \leq X^{-\sigma_M+o(1)} \mathcal{D}_{\text{ref}}(\zeta) \quad \text{for every declared } \zeta, \quad (227)$$

while the distinguished version is asserted only at $\zeta = \zeta_*(X)$. The harmonic theorem reduces this to a small $\mathfrak{h}_m(E_m)$ and a lower H_m^{eff} , but proves neither input. One omitted small cofactor can destroy a cardinal-density argument.

Door C. *Establish the represented reference in the same scope.* Prove

$$\|A_X \zeta\|^2 \geq X^{-\lambda_{\text{rep}}+o(1)} \mathcal{D}_{\text{ref}}(\zeta). \quad (228)$$

This step cannot absorb unproved critical-window occupancy, terminal activation, singleton survival, native erasure, or affine transversality into one unnamed exponent. Their established losses must be stated before λ_{rep} is used.

Door D. *Estimate the parity kernel, uniformly or pointwise.* The coefficient-uniform target is

$$\kappa_M = 1 + \lambda_{\max}(\Xi_{\neq p}) \leq X^{\nu_M+o(1)}. \quad (229)$$

It may be attacked through the two-affine formula (144), the determinant form (151), or the exact collision certificates. Pairwise correlation estimates must be upgraded to a Schur, block, stable-rank, or direct spectral bound. For the distinguished route, replace κ_M only by the exact Rayleigh quotient $\kappa_M(\zeta_*)$.

Door E. *Apply the strict effective-product test.* The zero-cost mass-degree door opens only when

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}}}. \quad (230)$$

At exponent equality one must prove the actual subpower margin $1 - \sqrt{q_X} \geq X^{-a+o(1)}$; it then costs $\lambda_{\text{reasm}} \leq 2a$. If the inequality fails, this certificate is silent. It does not prove that signed reassembly fails.

Door F. *Retain the shorted-Gram and signed fallbacks.* Independently test the TPC-60 uniform kernel condition

$$\text{Ker}(A_X + M_X) \subseteq \text{Ker } A_X. \quad (231)$$

An exact violation kills the uniform reassembly route. If it is killed, the physical distinguished vector may still survive through a direct estimate of

$$2 \text{Re}\langle A_X \zeta_*, M_X \zeta_* \rangle + \|M_X \zeta_*\|^2. \quad (232)$$

This fallback must not be relabeled coefficient-uniform.

Door G. *Close the complete endpoint ledger.* The final loss remains

$$\begin{aligned}\Lambda_{\text{tot}} := & \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} \\ & + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}},\end{aligned}\tag{233}$$

and the endpoint condition is

$$\boxed{\Lambda_{\text{tot}} < \frac{1}{400}}.\tag{234}$$

The exponents σ_M and ν_M are inputs to the reassembly test, not two additional losses to be double-counted. Paying the full coefficient-blind native degree is already closed by the 67/200 deficit; changing the normalization does not reopen it.

Conclusion. TPC-60 reduced missing-high restoration to a shorted-Gram decision tree. TPC-61 resolves three structural subquestions in that decision tree at the L0/L1 level. It identifies the exact realized cofactor defect, proves explicit one-dimensional cofactor-ladder and Brun-Titchmarsh harmonic certificates without assuming cancellation, and replaces the ambient native degree by the optimal restricted Gram. The remaining unequal-prime alias is now the explicit fixed- h_0 two-affine Möbius kernel (144), equivalently the signed determinant sum (151). This is a strict narrowing of the route, not a parity breakthrough. The next arithmetic theorem must control the actual retained-relation exposure and then beat the normalized parity-kernel exponent in the same scope.

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